

Module 3

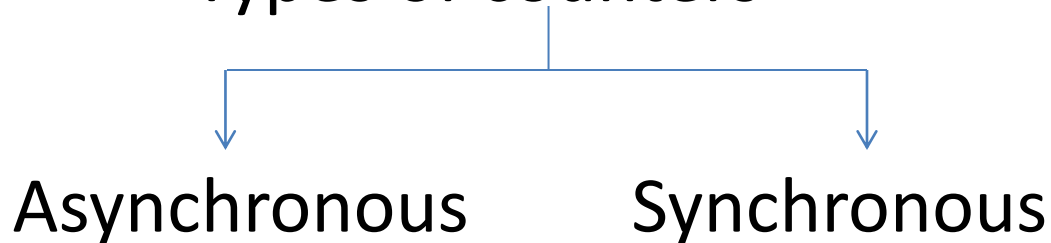
Sequential Circuits

3.2_ Counters

Counters

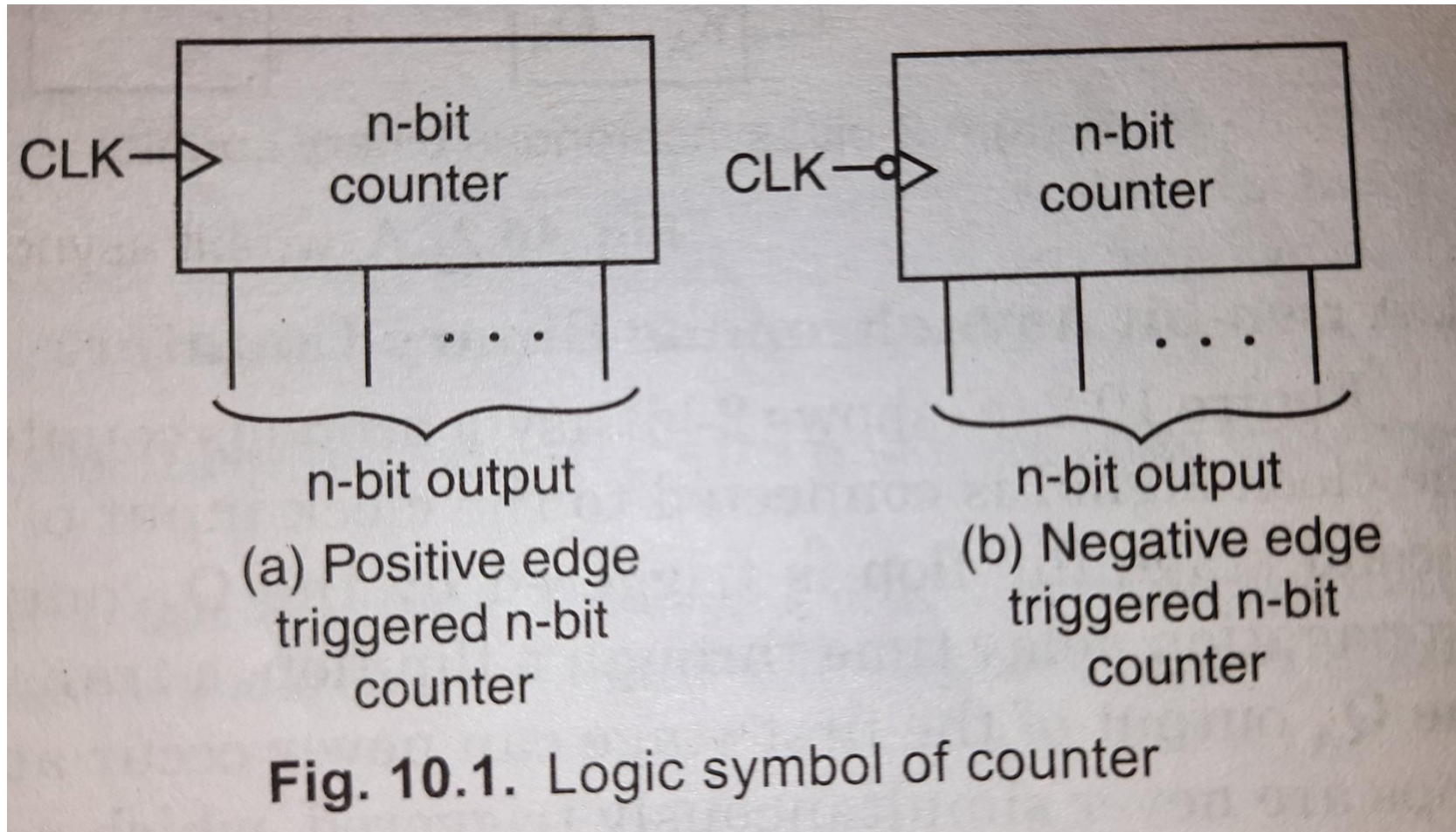
- A counter is register capable of counting the number of clock pulses arriving at its clock input
- Counter ckt is designed using Flip Flops

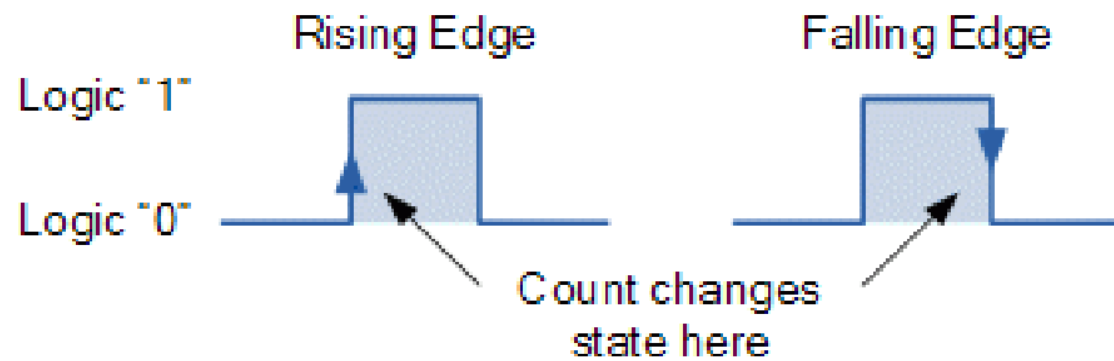
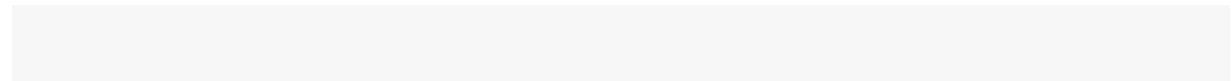
Types of counters



- Up, Down, Up/Down
- Decade
- Mod-n (Modulus)
- Ring counter

Logic symbol of counter





OVERVIEW OF COUNTERS

- **Counter-by definition**
 - One input (clock)
 - Outputs follow defined sequence
- **Common tasks of counter**
 - Count up or down
 - Increment or decrement count
 - Sequence events
 - Divide frequency
 - Address memory
 - As memory

Ripple Counter

- Ripple counter is an Asynchronous counter. It got its name because the clock pulse ripples through the circuit.
- An n-MOD ripple counter contains n number of flip-flops and the circuit can count up to 2^n values before it resets itself to the initial value.
- Count will be from 0 to $2^n - 1$

CHARACTERISTICS OF COUNTERS

- **Number of bits (4-bit, 8-bit, etc.)**
- **Maximum count**
 - 4 bit = $2^4 = 0000$ to 1111 in binary
 - 8 bit = $2^8 = 0000\ 0000$ to $1111\ 1111$ in binary
- **Modulus of counter-number of states**
 - Decade counter
 - 4-bit
 - 8-bit
 - Mod-n
- **Up or down counter**
- **Asynchronous or synchronous counter**
- **Presettable counter**
- **Self-stopping counter**

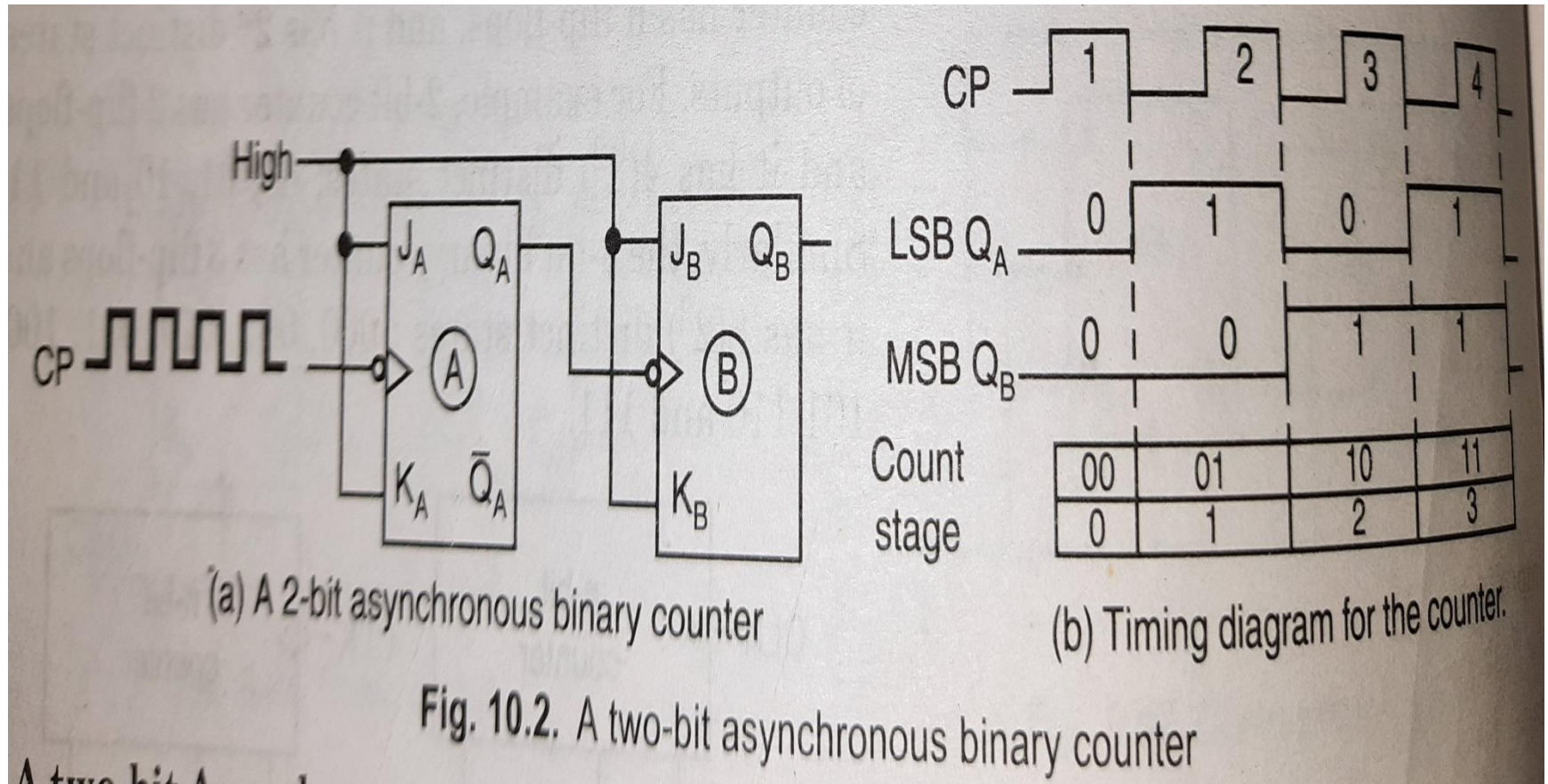
Asynchronous counter

- An Asynchronous counter can count using **Asynchronous clock input**. Counters can be easily made using [flip-flops](#).
- As the count depends on the clock signal, in case of an Asynchronous counter, changing state bits are provided as the clock signal to the subsequent flip-flops.
- Those Flip-flops are serially connected together, and the clock pulse ripples through the counter.
- Due to the ripple clock pulse, it's often called a ripple counter.
- An Asynchronous counter can count $2^n - 1$ possible counting states.

Clocking condition for up and down counter

Negative edge triggered clock	Positive edge triggered clock
UP Counter -: Q is clock for next FF	UP Counter -: Q' is clock for next FF
DOWN Counter -: Q' is clock for next FF	DOWN Counter -: Q is clock for next FF

Two bit Asynchronous up counter



Three bit Asynchronous up counter

3. Timing Diagram

Figure 10.2 (b) shows the timing diagram for two-bit asynchronous counter. It illustrates the changes in the state of the flip-flop outputs in response to the clock. J and K input of JK flip-flops are tied to logic HIGH hence output will toggle for each negative edge of the clock input.

EXAMPLE 10.3. Extend the counter shown in figure 10.2 for 3-stages, and draw output waveforms.

Solution:

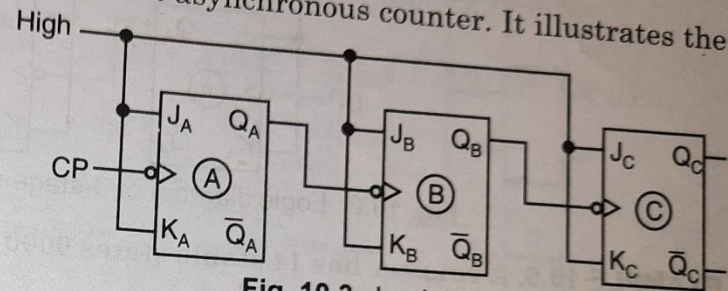


Fig. 10.3. Logic diagram

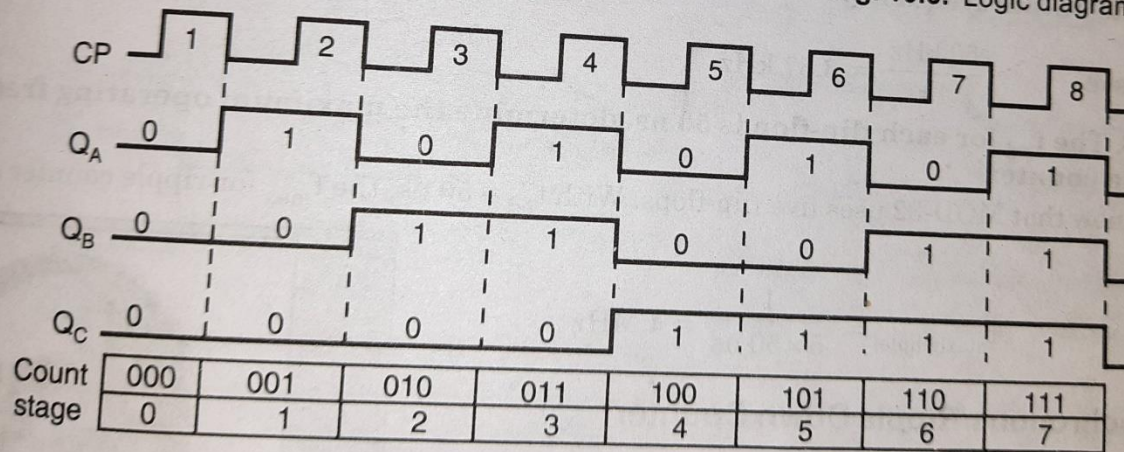
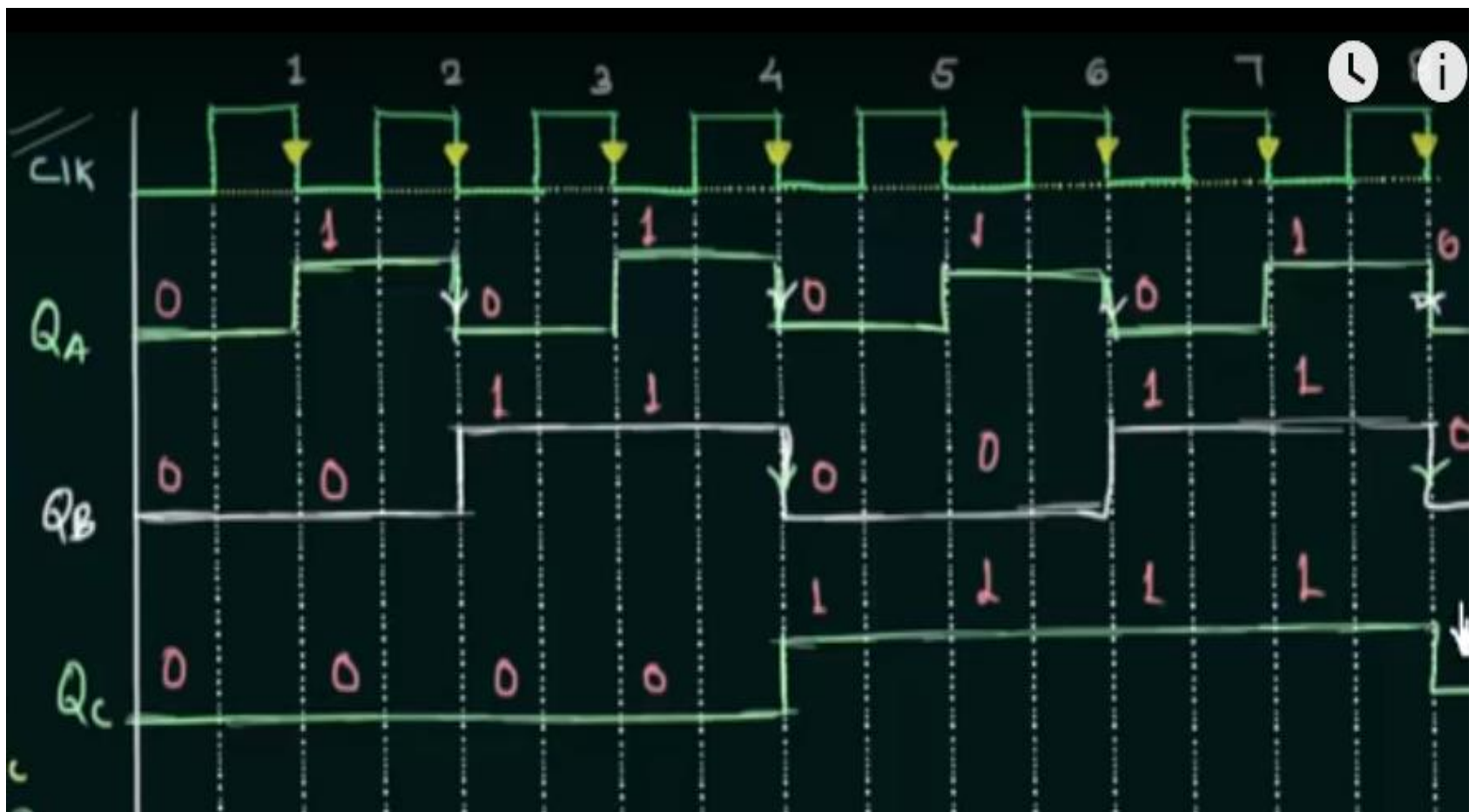


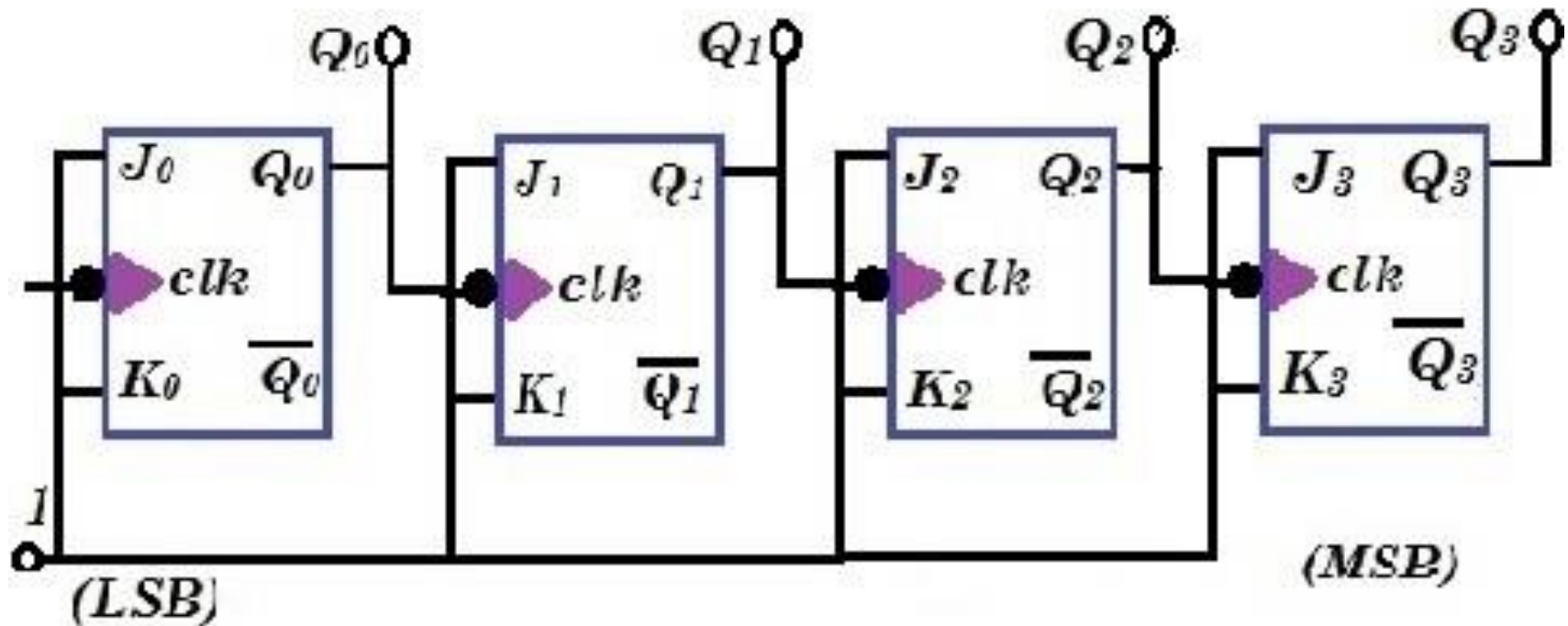
Fig. 10.4. Output waveforms for 3-bit asynchronous counter.

Truth Table of 3 bit Asynchronous up counter

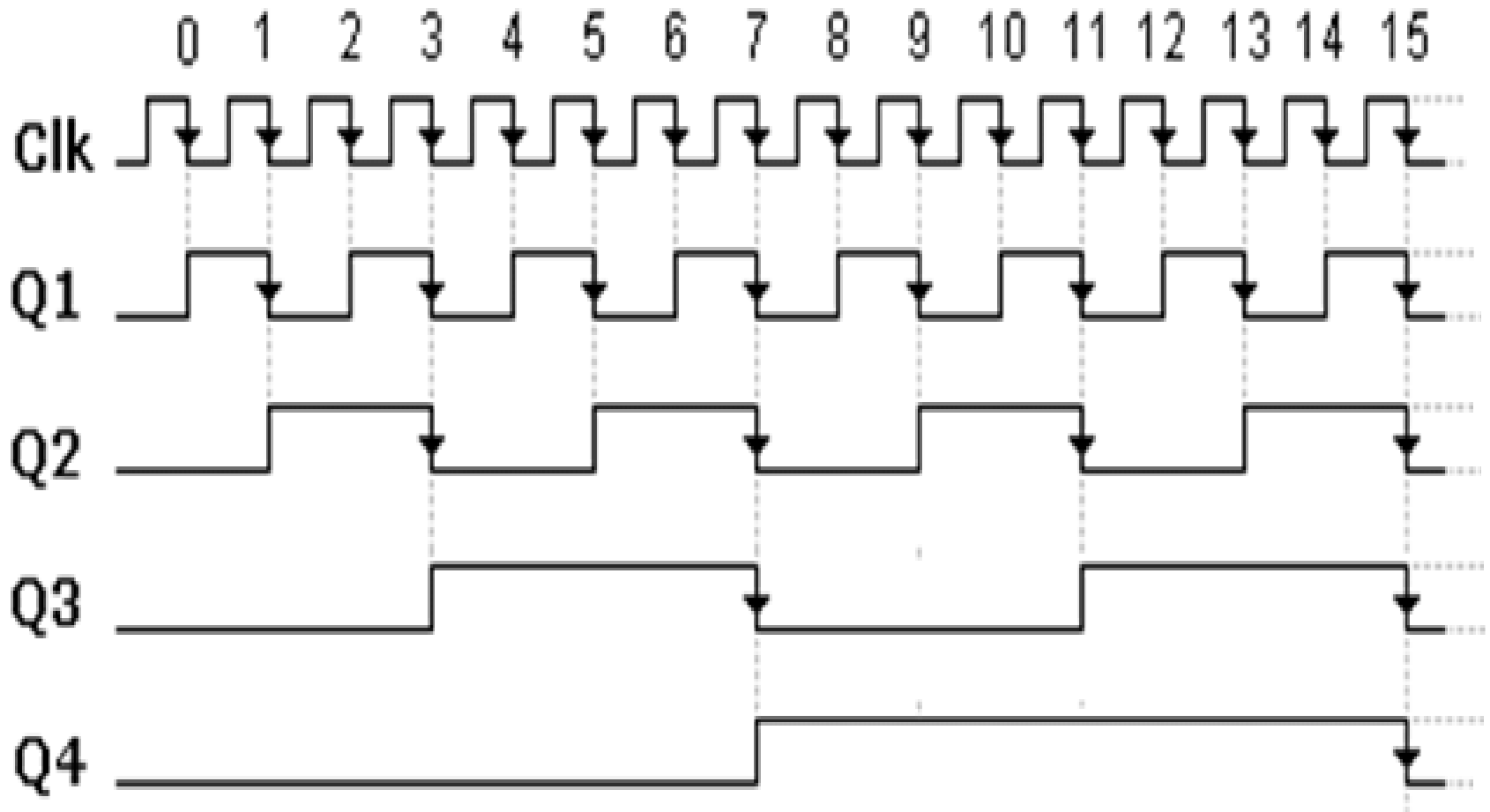
No of negative edge of Clock	Q_A (LSB)	Q_B	Q_C (MSB)
0	0	0	0
1	1	0	0
2	0	1	0
3	1	1	0
4	0	0	1
5	1	0	1
6	0	1	1
7	1	1	1



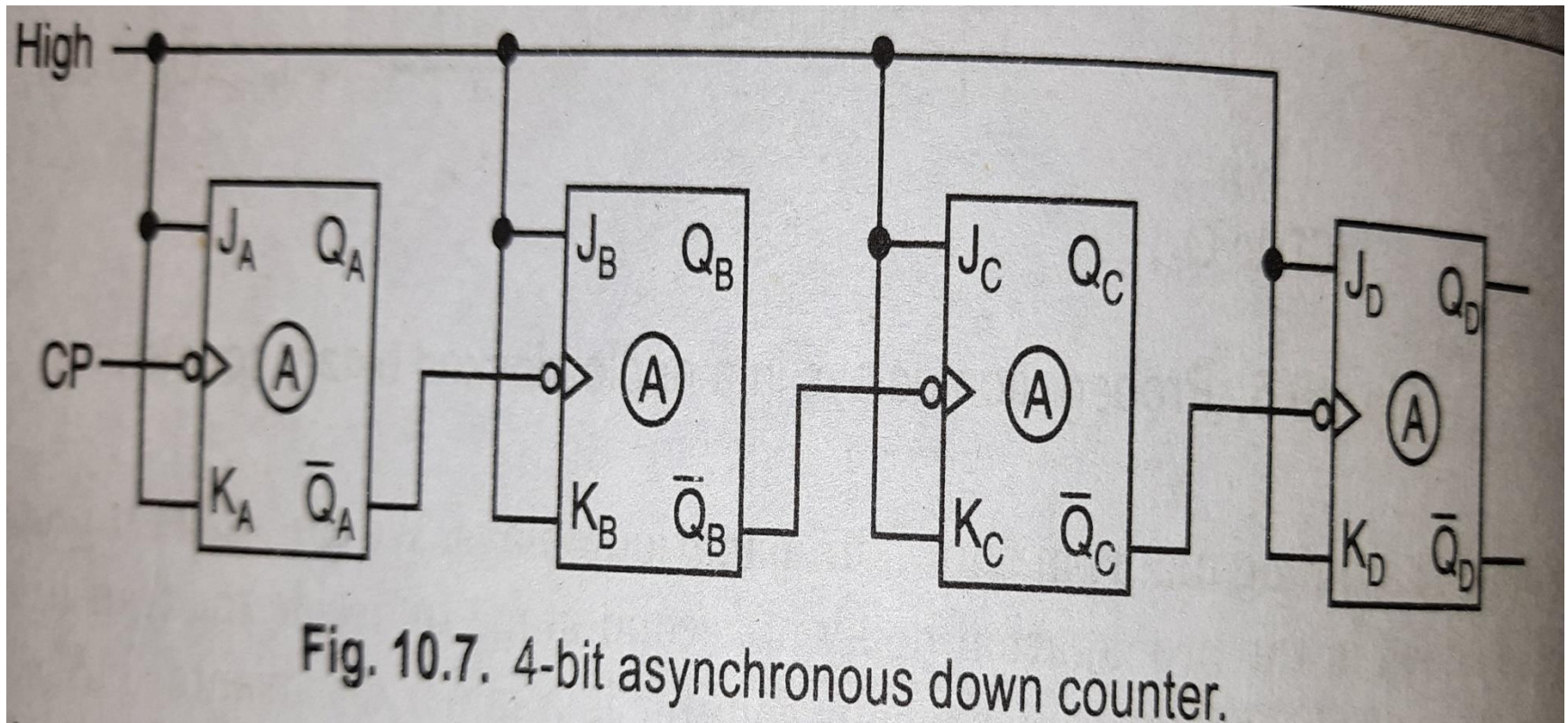
4 bit Asynchronous Up counter



Timing diagram for 4 bit Asynchronous up counter



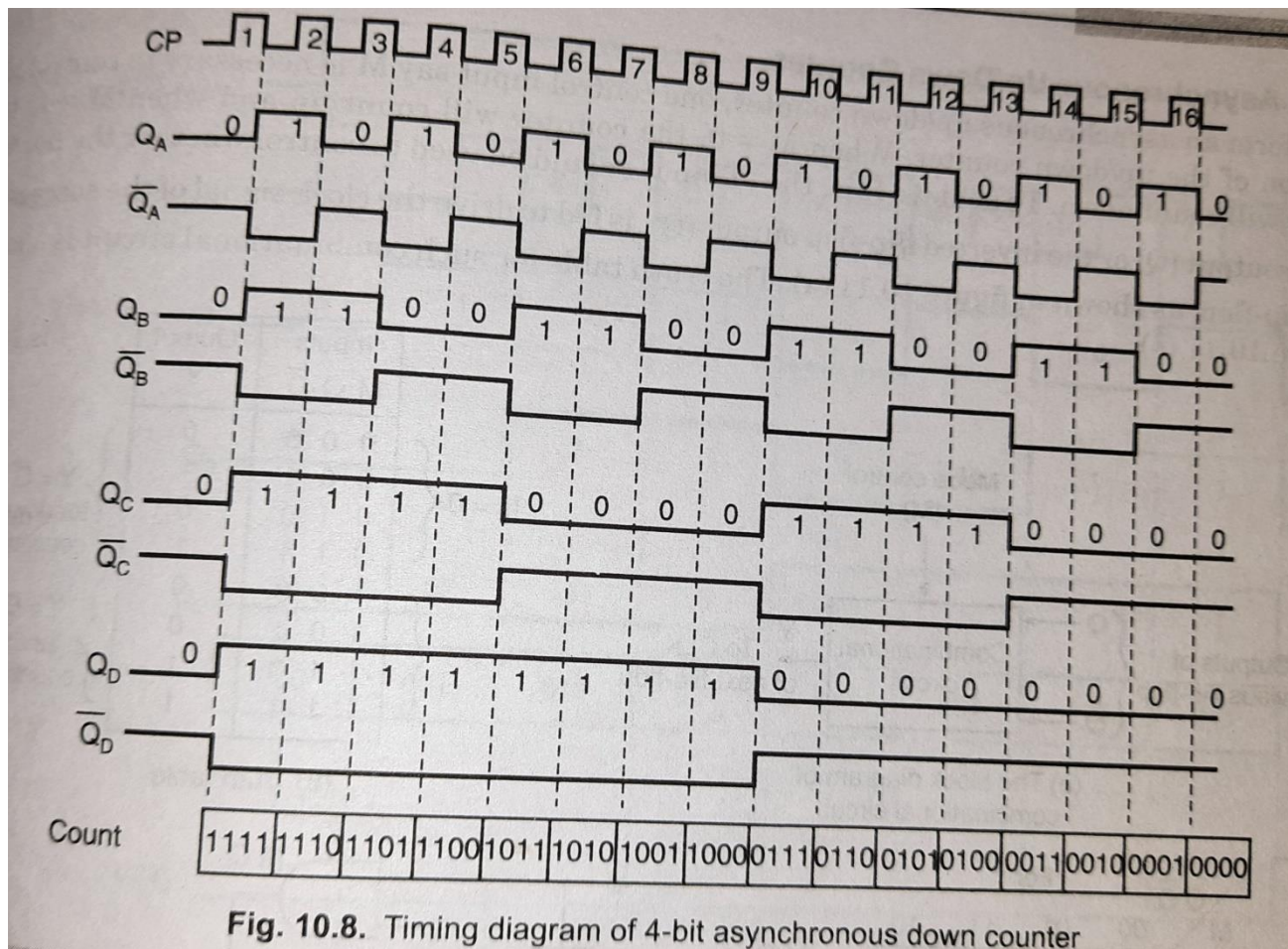
Four bit Asynchronous down counter



Working of 4 bit Asynchronous down counter

- First flip flop with negative edge clock
- Q'_A output of first FF is connected as clock input of FF_B
- When Q'_A changes from high to low, at that falling edge FF_B is triggered
- Q'_B output of second FF_B is connected as clock input of FF_C and so on

Waveforms four bit Asynchronous down counter



REVIEW

- An “up” counter may be made by connecting the clock inputs of positive-edge triggered J-K flip-flops to the Q' outputs of the preceding flip-flops.
- Another way is to use negative-edge triggered flip-flops, connecting the clock inputs to the Q outputs of the preceding flip-flops.
- In either case, the J and K inputs of all flip-flops are connected to V_{cc} or V_{dd} so as to always be “high.”
- Counter circuits made from cascaded J-K flip-flops where each clock input receives its pulses from the output of the previous flip-flop invariably exhibit a *ripple effect*, where false output counts are generated between some steps of the count sequence. These types of counter circuits are called *asynchronous counters*, or *ripple counters*.

Disadvantage of Asynchronous Counter Circuit: Limited Speed

- Main disadvantage of the asynchronous, or ripple, counter circuit is limited speed.
- While all gate circuits are limited in terms of maximum signal frequency, the design of asynchronous counter circuits compounds this problem by making propagation delays additive.

Propagation delays in Asynchronous counter

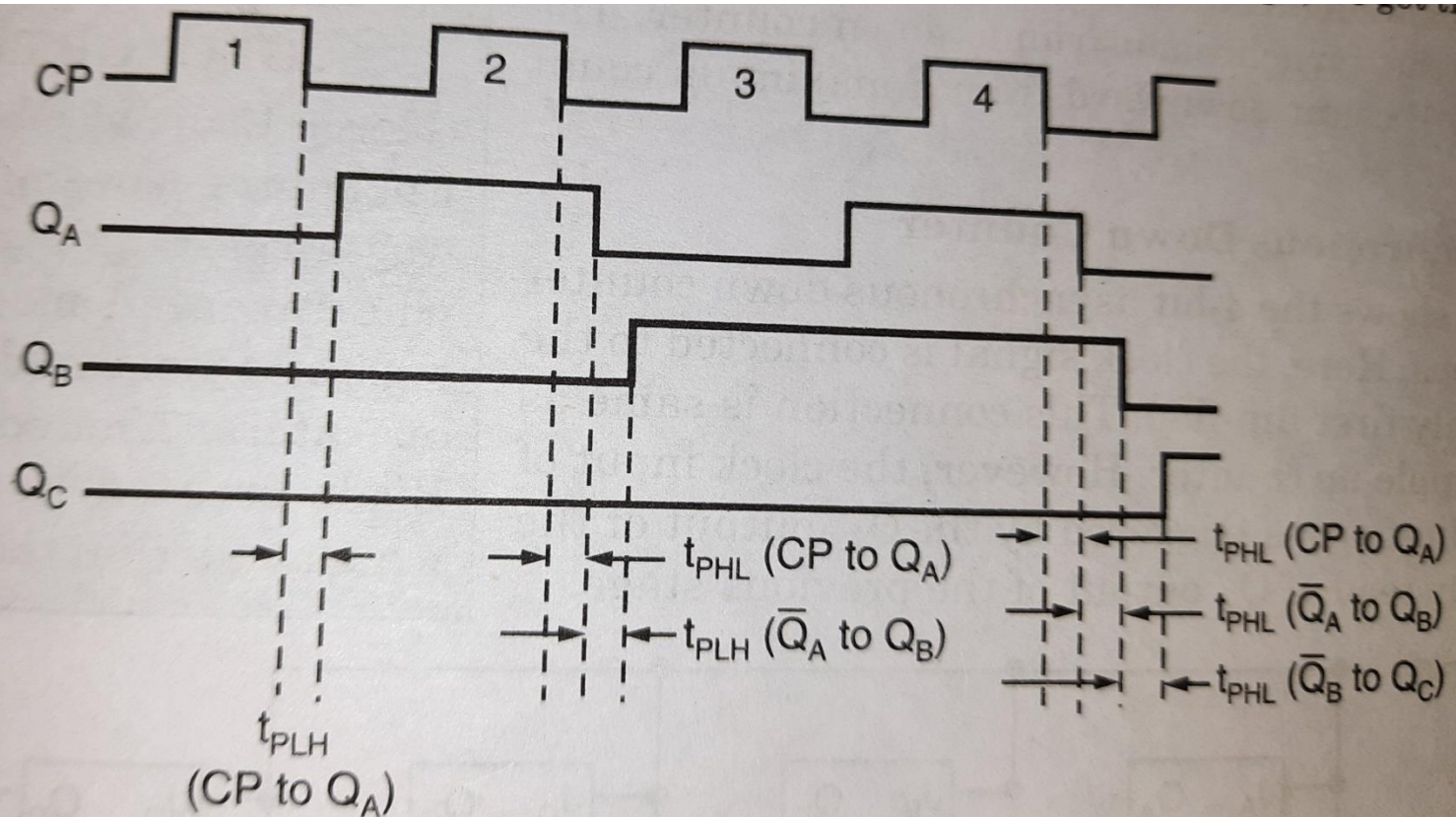


Fig. 10.5. Propagation delays in a ripple clocked binary counter.

3 bit Asynchronous Up/Down counter

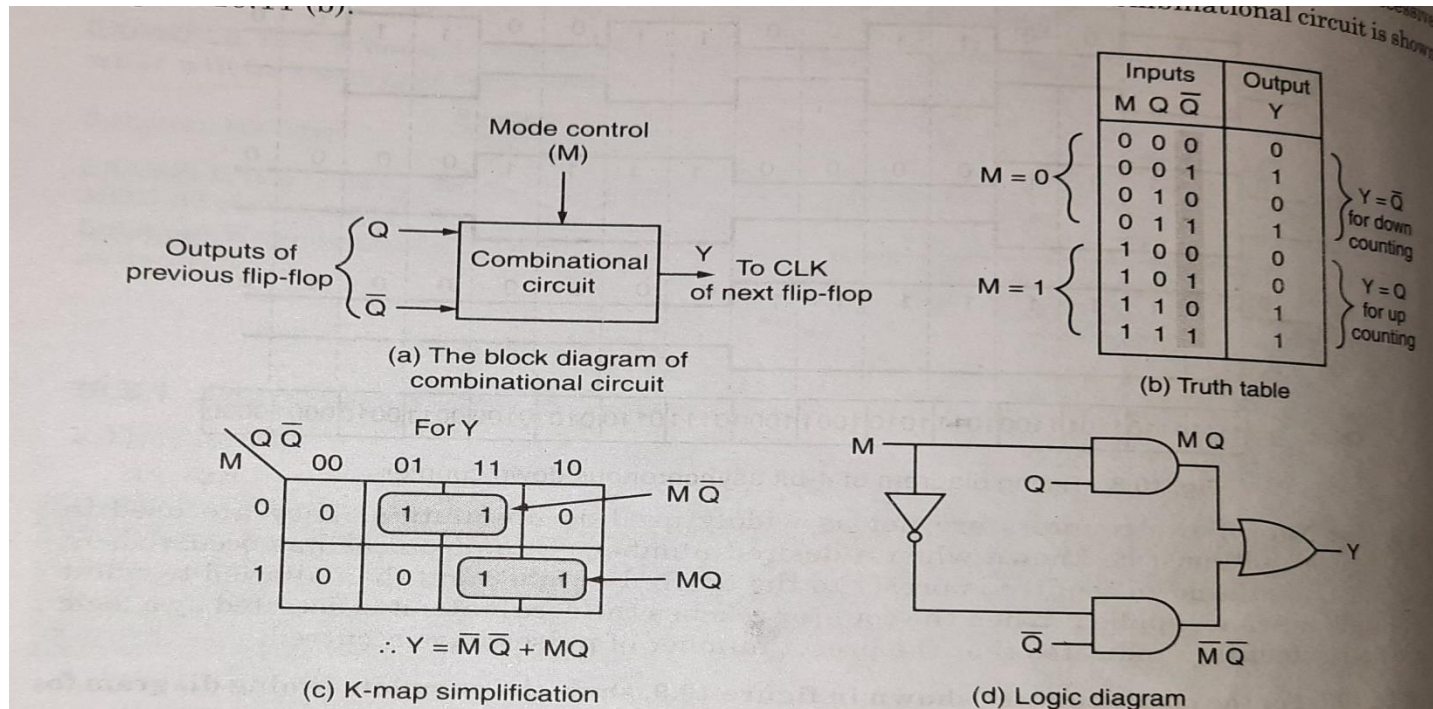


Fig. 10.11.

Figure 10.12 shows the 3-bit up/down counter that will count from 000 up to 111 when the mode control input M is 1 and from 111 down to 000 when mode control input M is 0.

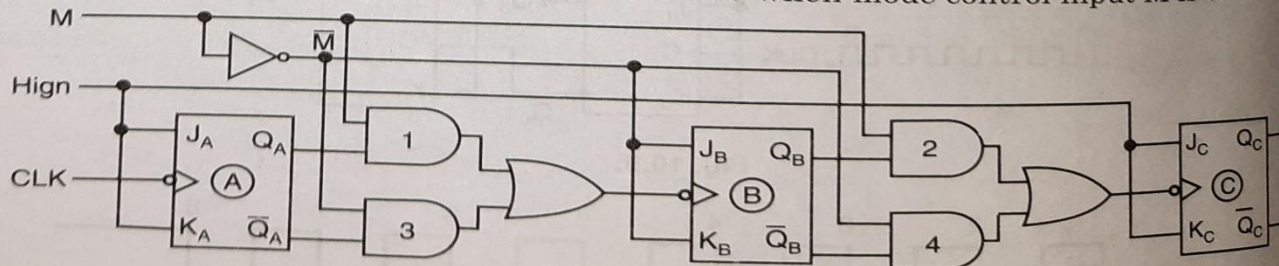
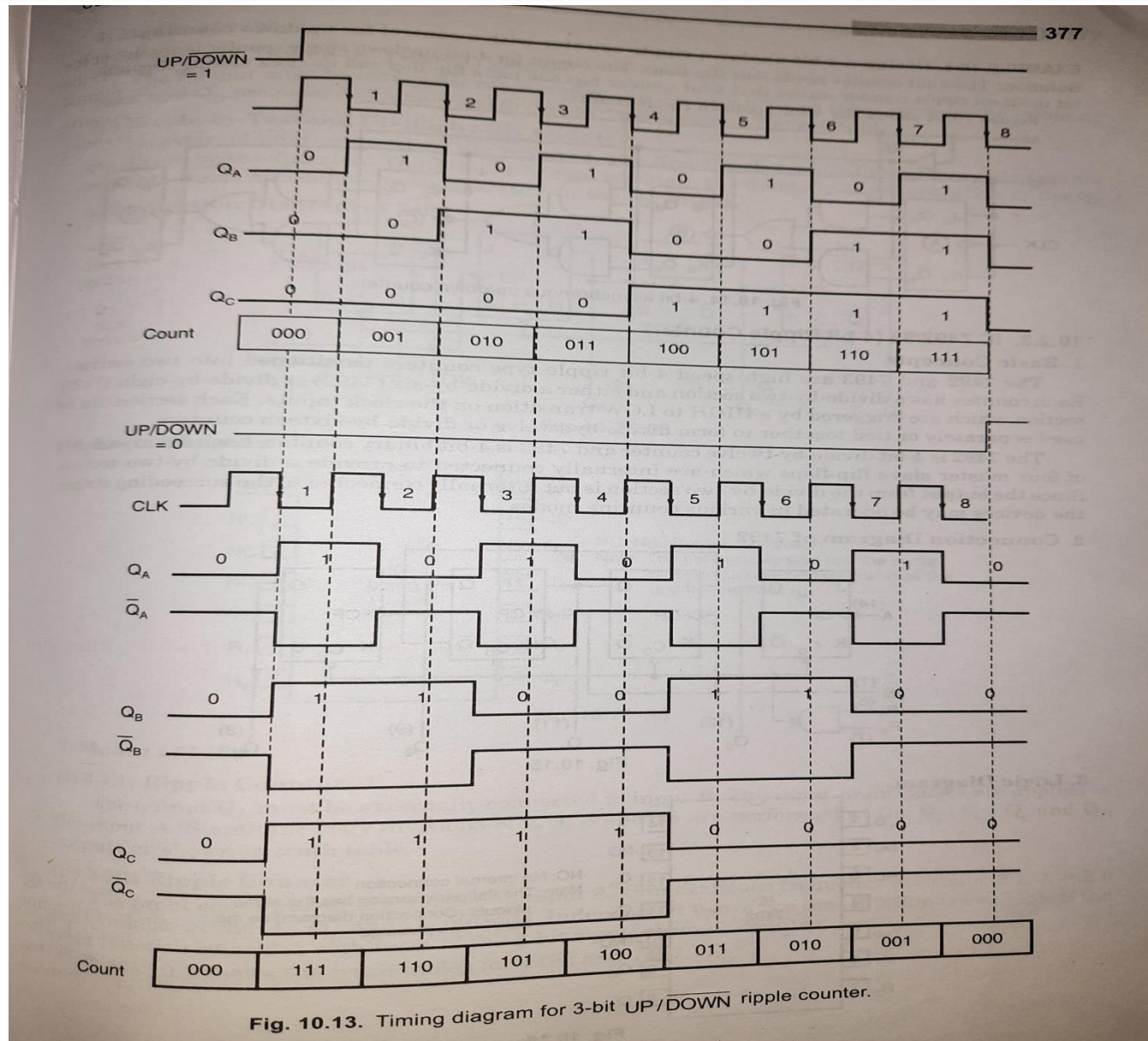


Fig. 10.12. 3-bit asynchronous up/down counter.

Timing diagram of Asynchronous Up/Down counter



10.3. DECODING GATES

Decoding gates are used to indicate whether counter has reached to particular state. The outputs of the counter are connected to the AND gate as inputs and the output of the AND gate goes high for particular state. Let us consider figure 10.21. Here, the output of decoding gate goes high when counter outputs are $C = 1$, $B = 1$ and $A = 1$. In figure 10.22, the output of decoding gate goes high when counter outputs are $C = 1$, $B = 0$ and $A = 0$.

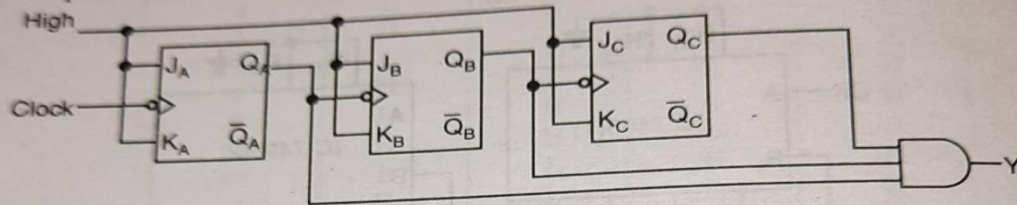


Fig. 10.21. Decoding gate to indicate state 7 (111)

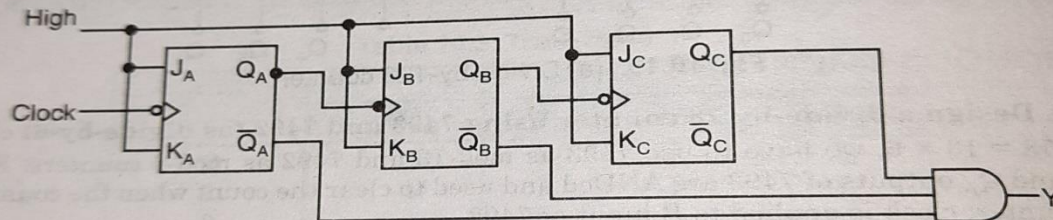


Fig. 10.22. Decoding gate to indicate state 4 (100)

Similarly, we can connect corresponding outputs to decoding gate inputs to indicate desired state. Figure 10.23 gives these connections for all possible state detection for 3-bit counter.

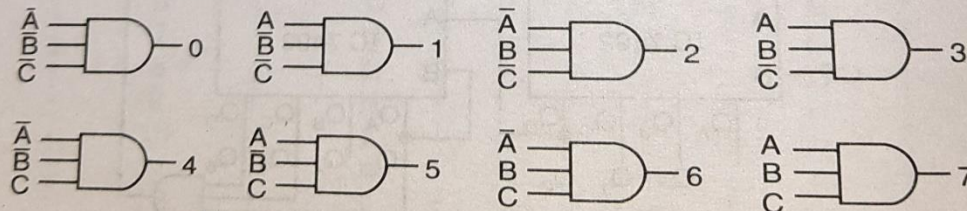


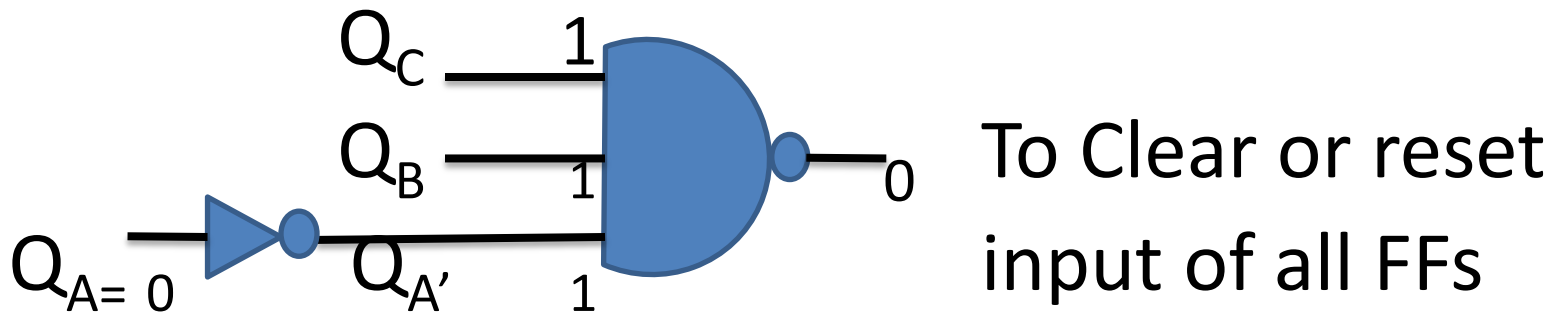
Fig. 10.23. Decoding gate connections.

Modulus-n (Mod-n) counter

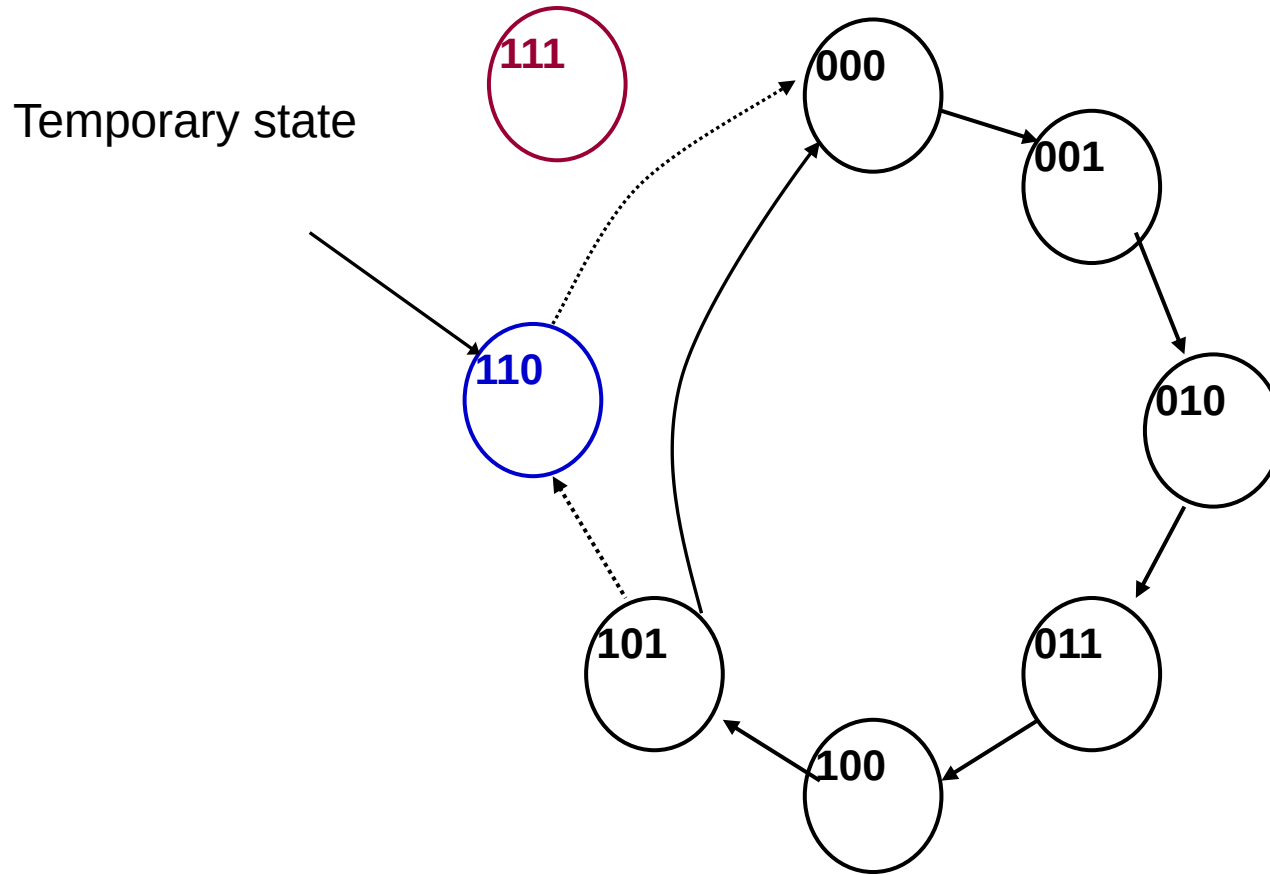
- For 2 bit ripple counter it is Mod-4 counter
- For 3 bit ripple counter it is Mod-8 counter
- If n = no of bits
- Mod number = 2^n
- State = 2^n
- Max count = $2^n - 1$
- Modulus of counter represents the no of states or gives how many states counter will have
- e.g. Mod-4 up counter will count as 00,01,10,11
- Mod-4 down counter will count as 11,10,01,00

Designing mod-n counter

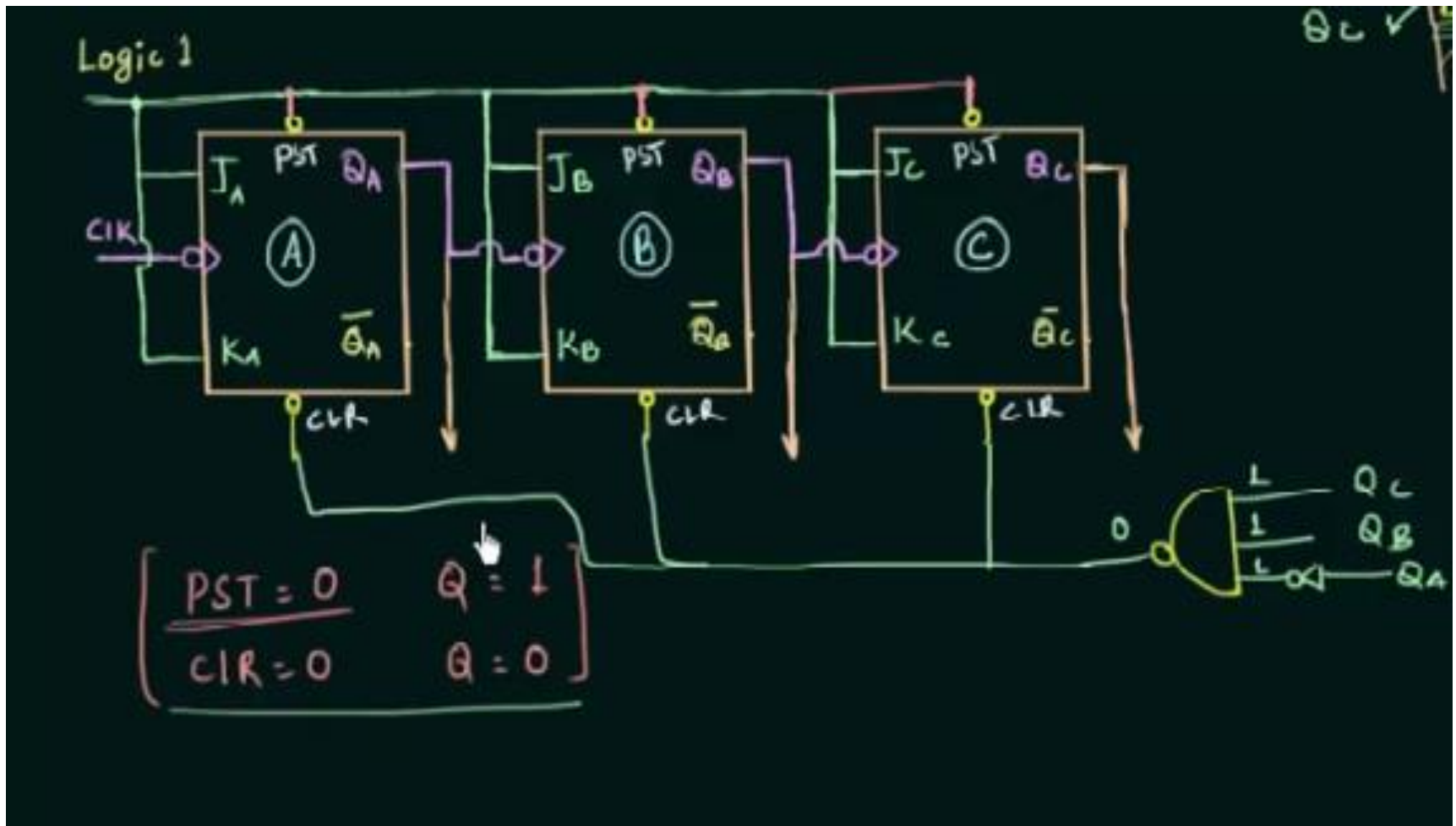
- Design Mod-6 counter(Up) using Mod-8 counter
- Total no of states = 6, Max count = 6-1 = 5
- Count will be 000,001,010,011,100,101
- As 101 count is over and when clock is arriving for 110 at this instant count should be 000. So, all FFs will be cleared to 000 using additional combinational control and using Clear/Reset input of FFs
- So as per 110, $Q_C=1$, $Q_B=1$ and $Q_A=0$



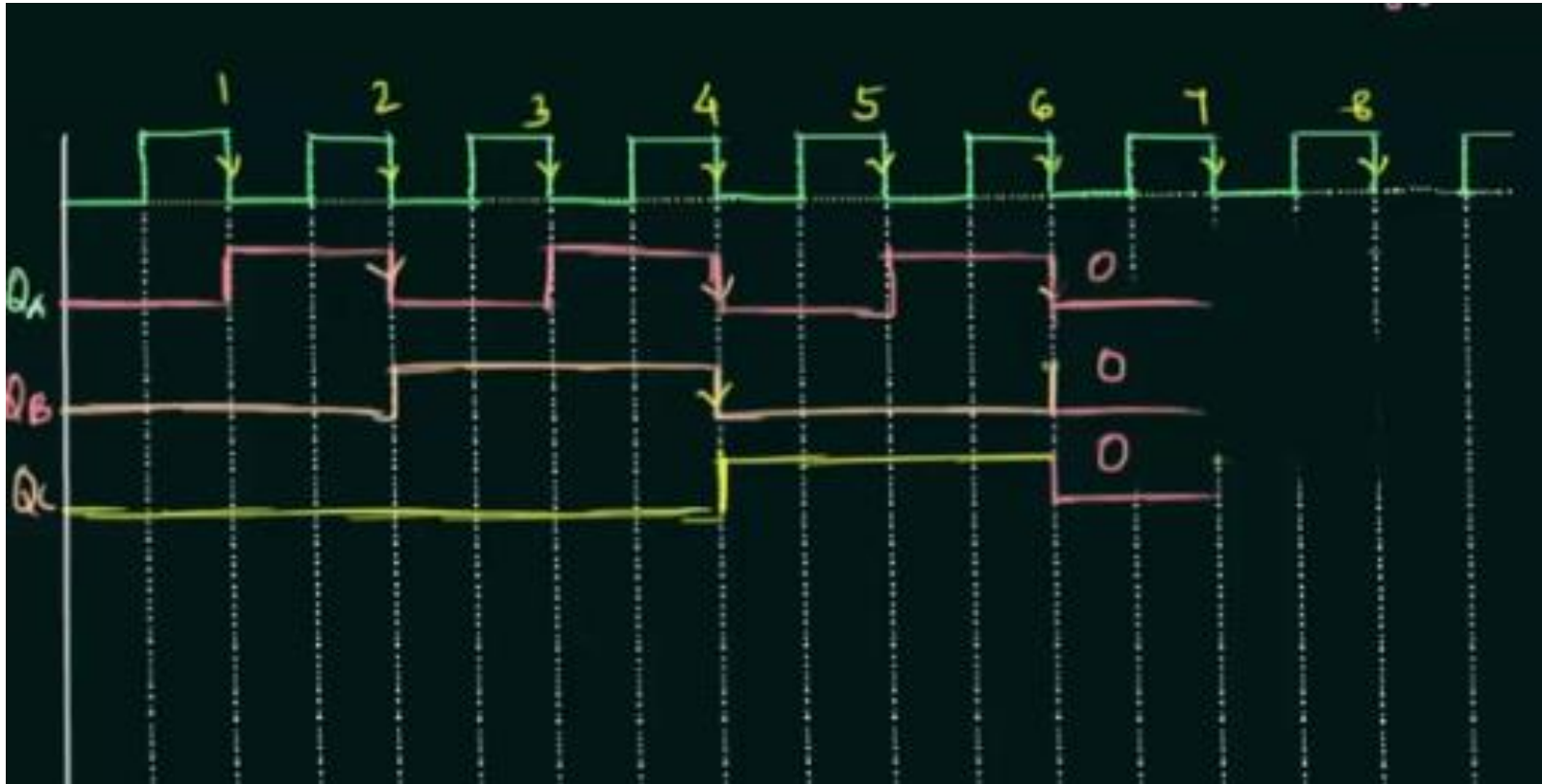
State diagram of Mod -6 Asynchronous up counter



Circuit diagram of Mod -6 Asynchronous up counter



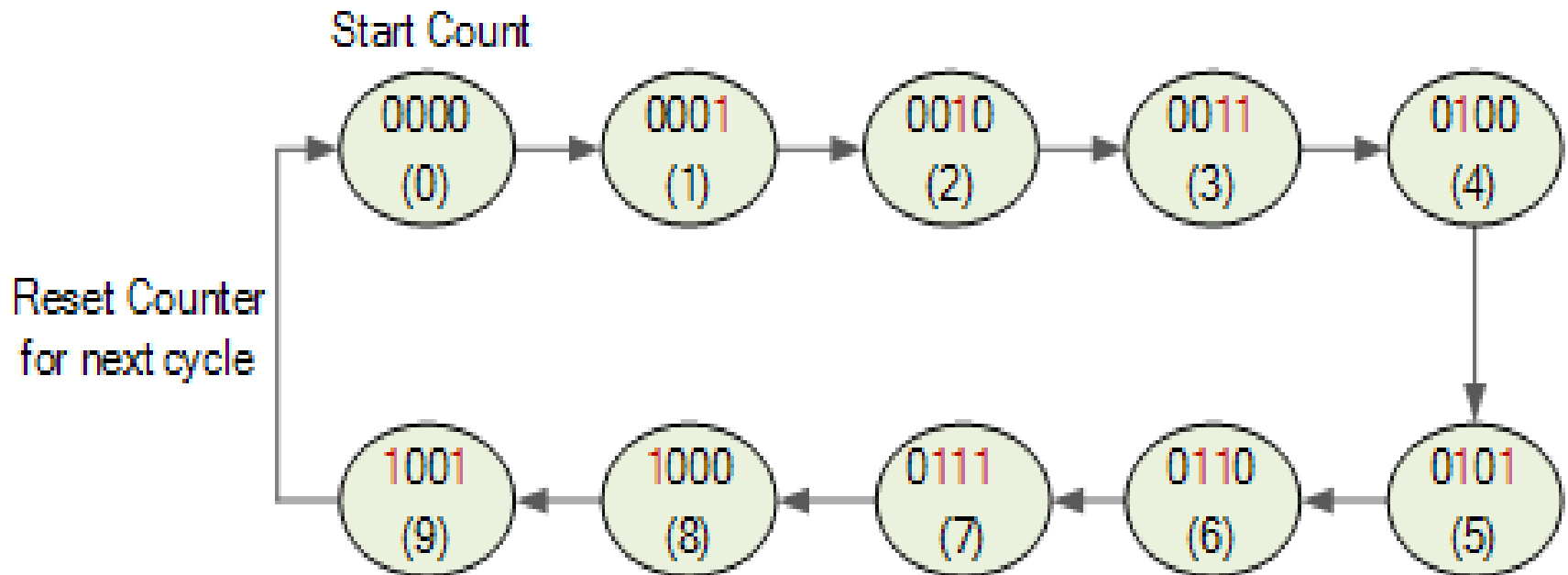
Timing diagram of Mod -6 Asynchronous up counter



Decade(BCD) ripple up counter

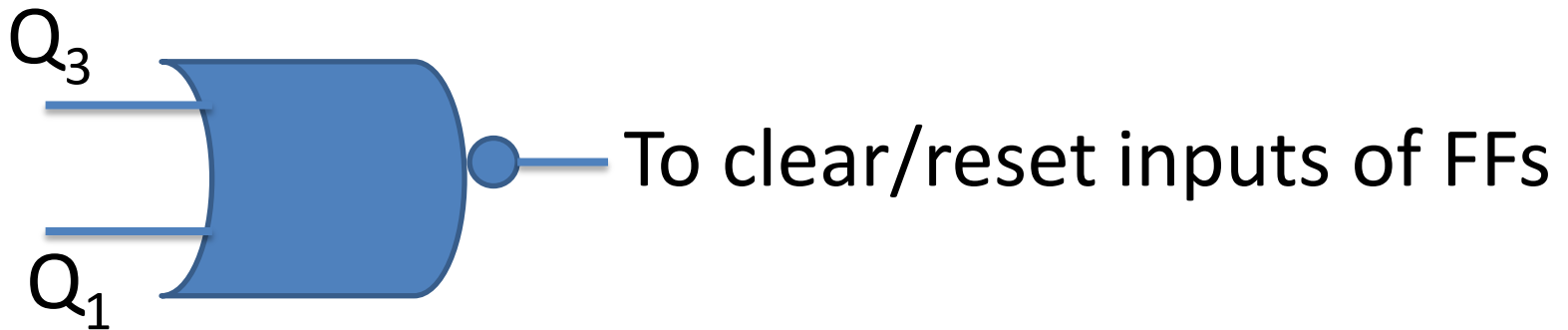
- No of states=10, Max count = 10-1=9
- Counting

0000,0001,0010,0011,0100,0101,0110,0111,
1000,1001 and then again start from 0000

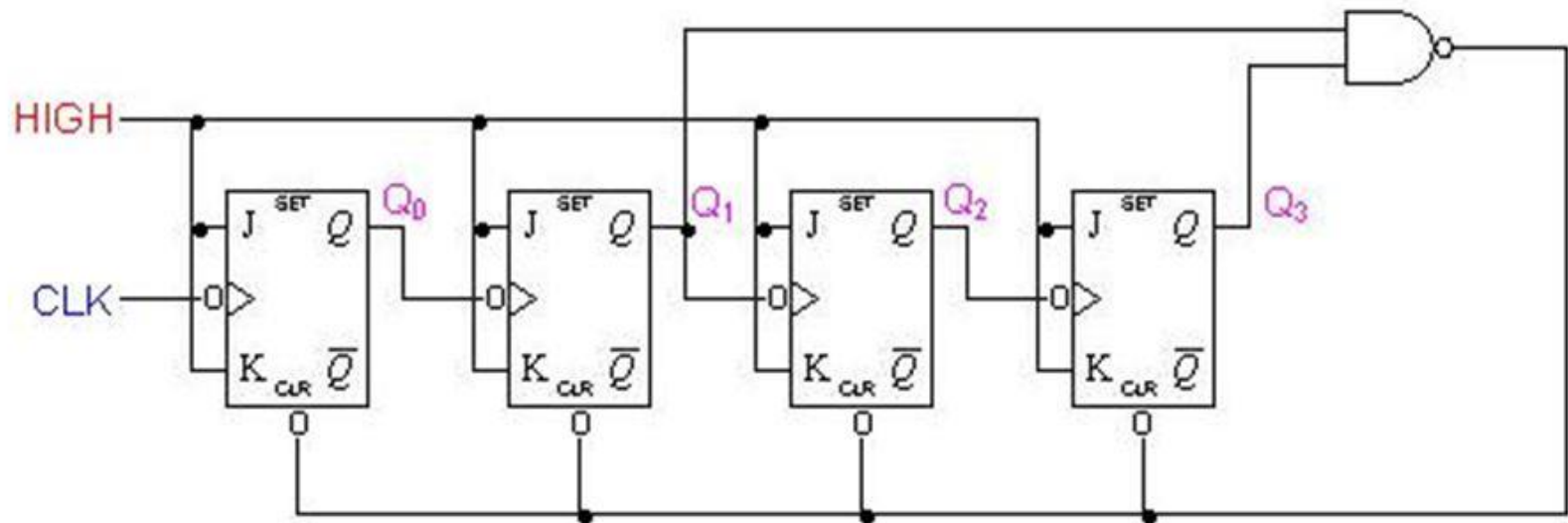


- As soon as 10th clock pulse arrives, counter should stop counting at 1001 and again start from 0000
- So accordingly FF outputs will be

$$Q_3 = 1, Q_2 = 0, Q_1 = 1, Q_0 = 0$$



A Decade Counter (asynchronous counter)



Synchronous Counter

- When counter is clocked such each ff in the counter is triggered at the same time, the counter is called as Synchronous counter
- the flip-flops are clocked at the same time by a common clock pulse
- Clock is common to all FFs

Design steps for Synchronous counter

- Step 1: Decide number of FFs
- Step 2: Prepare excitation table of FF
- Step 3: i) Prepare state diagram
and ii) circuit excitation table
- Step 4: Obtain simplified equations using K map with reference of circuit excitation table
- Step 5: Draw circuit diagram from Boolean equations derived using K map

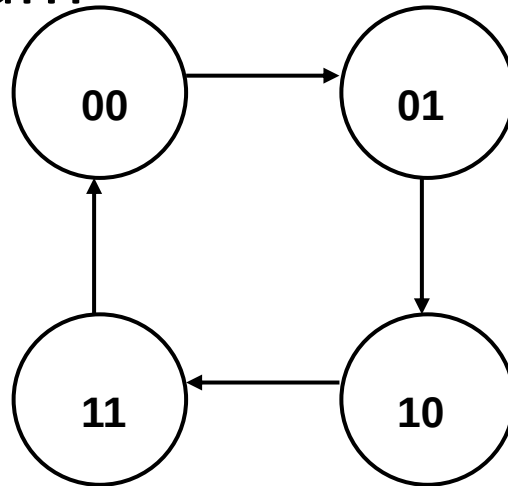
Design of 2 bit Synchronous up counter

- Step 1: For 2 bits FFs required =2, JK FF
- Step 2: Prepare Excitation table of JK FF

Present State Q_n	Next State Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Design of 2 bit Synchronous up counter

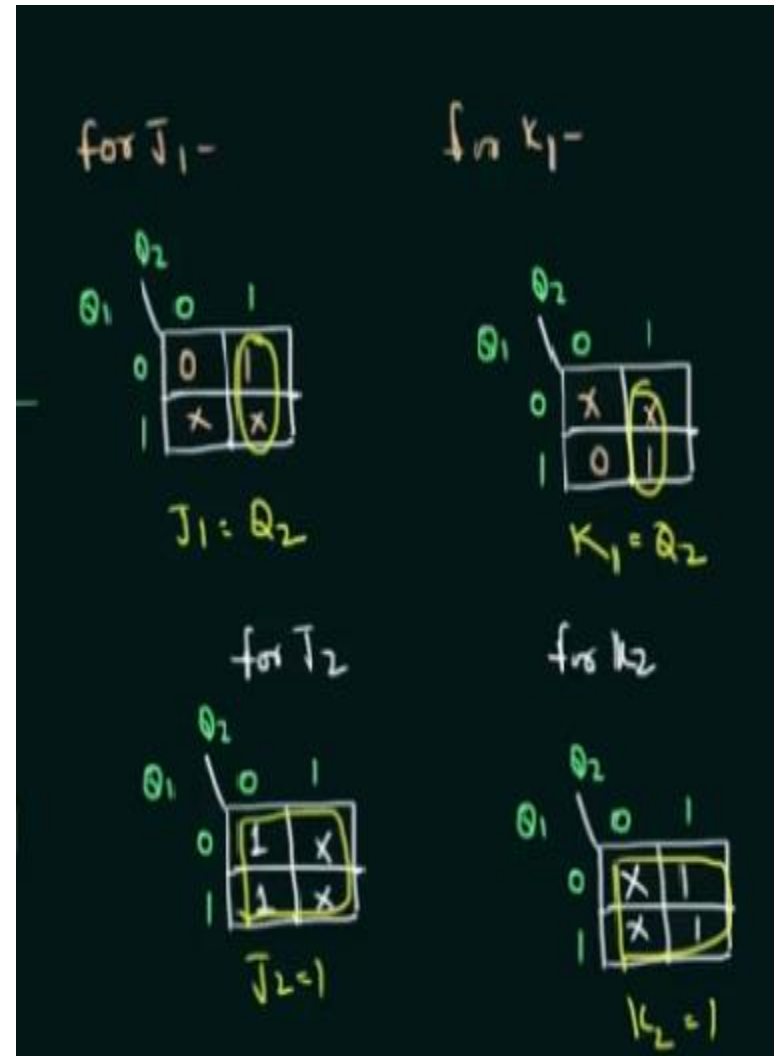
- Step 3: i) State Diagram
- As $n=2$, Count $2^n = 4$, Max count = $2^n - 1 = 3$
- Count 00->01->10->11->00...
- State Diagram



Step3:ii)Circuit excitation table

Present State Q_n		Next State Q_{n+1}		JK inputs for FF1		JK inputs for FF2	
Q_1	Q_2	Q_1 +1	Q_2 +1	J1	K1	J2	K2
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

Step4: K map from circuit excitation table

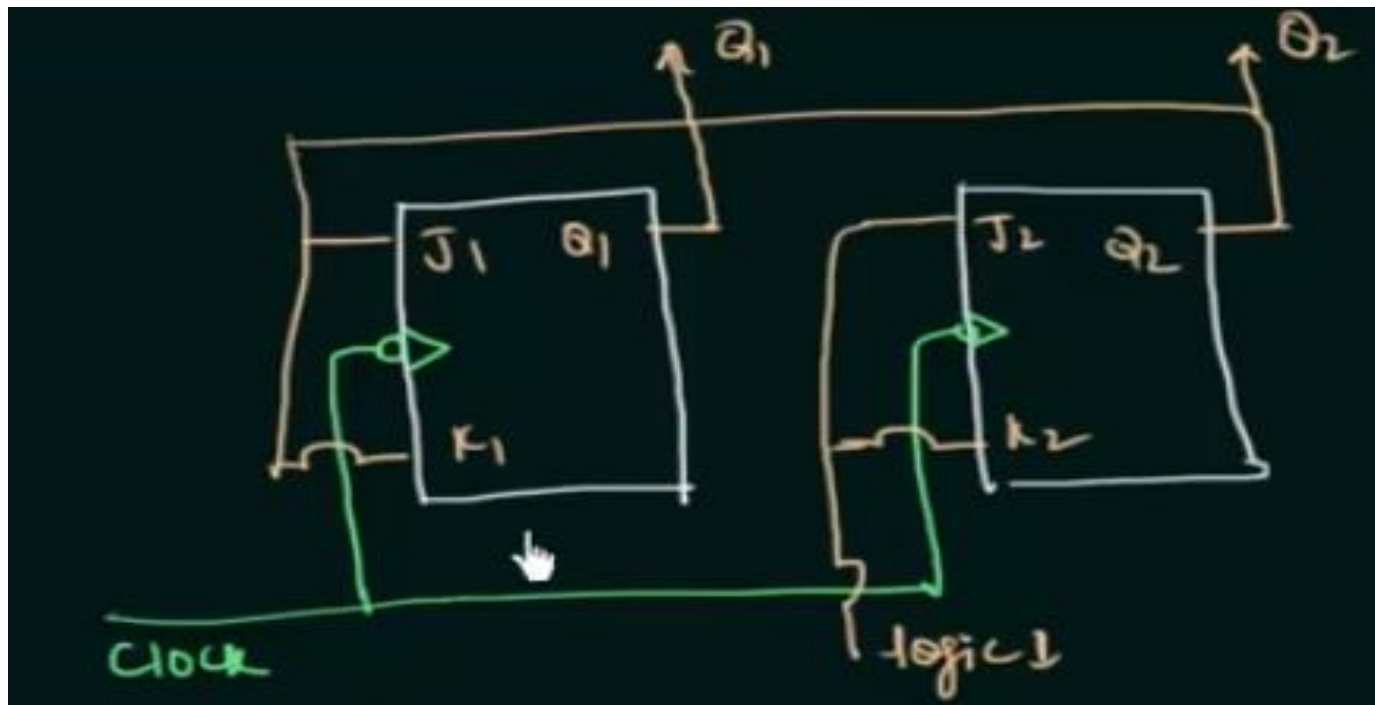


Step 5: Circuit diagram of 2 bit Synchronous up counter

From K map

$$J_1 = K_1 = Q_2$$

$$J_2 = K_2 = 1$$



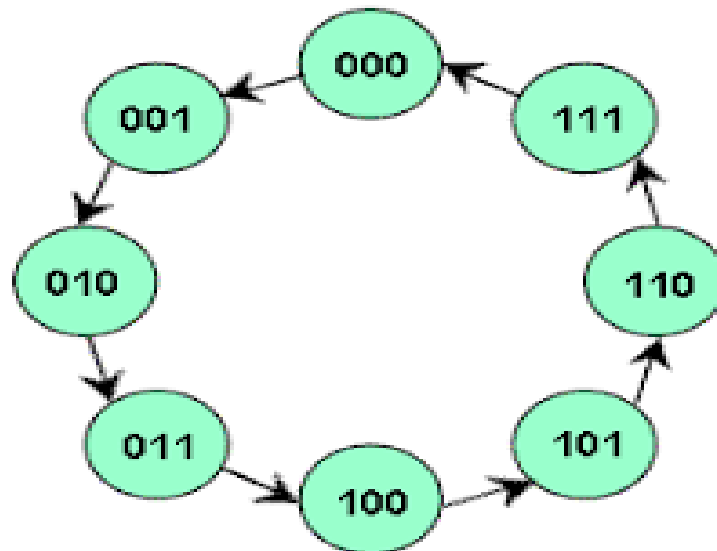
Design of 3 bit Synchronous up counter

- Step 1: Determine No of FFs required = 3, Type of FF-T FF
- Step 2: Prepare excitation table of T FF

Present State	Next State	Input
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Design of 3 bit Synchronous up counter contd..

- Step 3:i) Prepare state diagram
- $N=3$, $2^3=8$ states, Max count= $2^3-1=7$
- 000->001->010->011->100->101->110->111



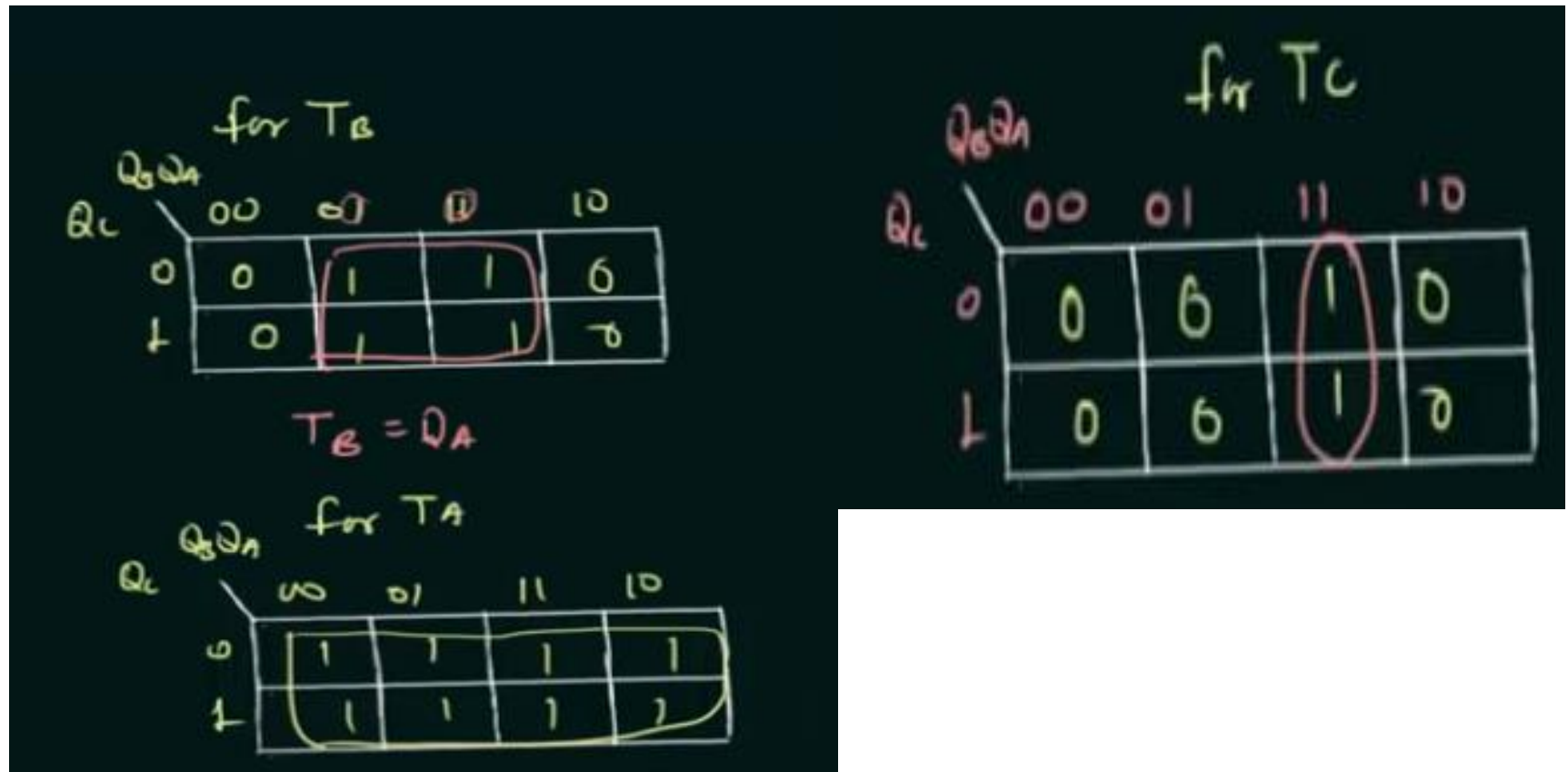
Design of 3 bit Synchronous up counter contd..

- Step 3:ii) Prepare circuit excitation table

Present State			Next State			Input		
Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	T_C	T_B	T_A
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

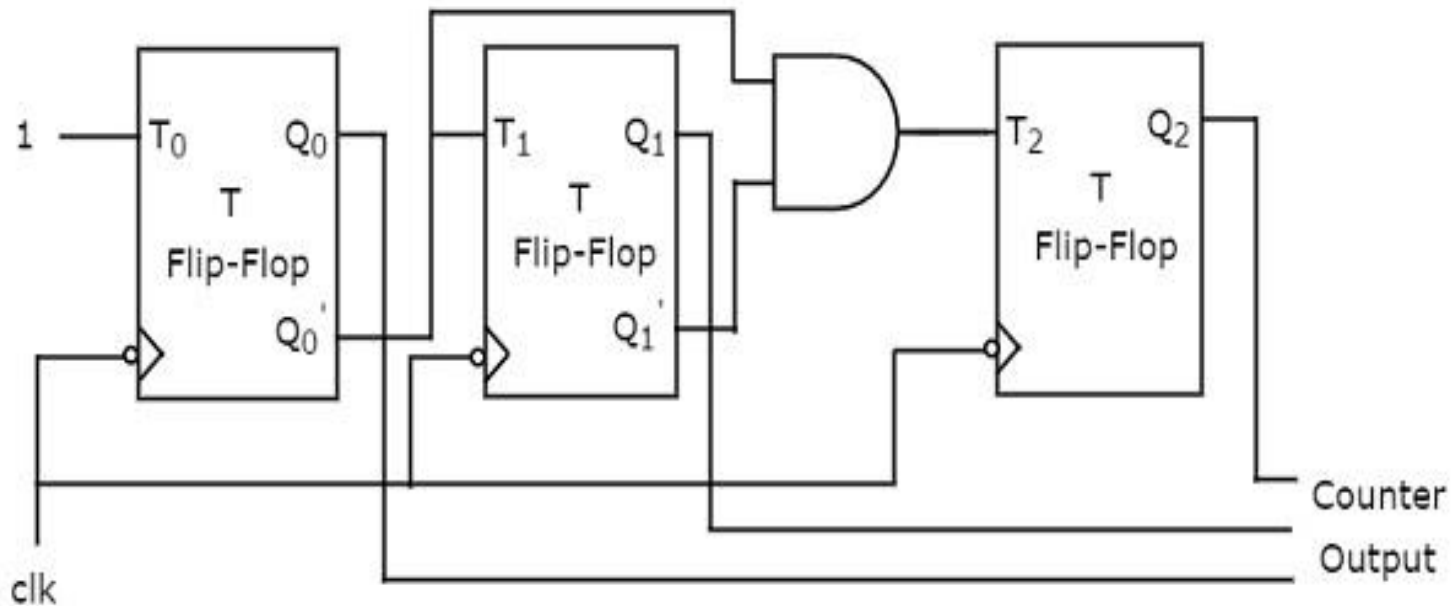
Design of 3 bit Synchronous up counter contd..

- Step 4: K map from circuit excitation table

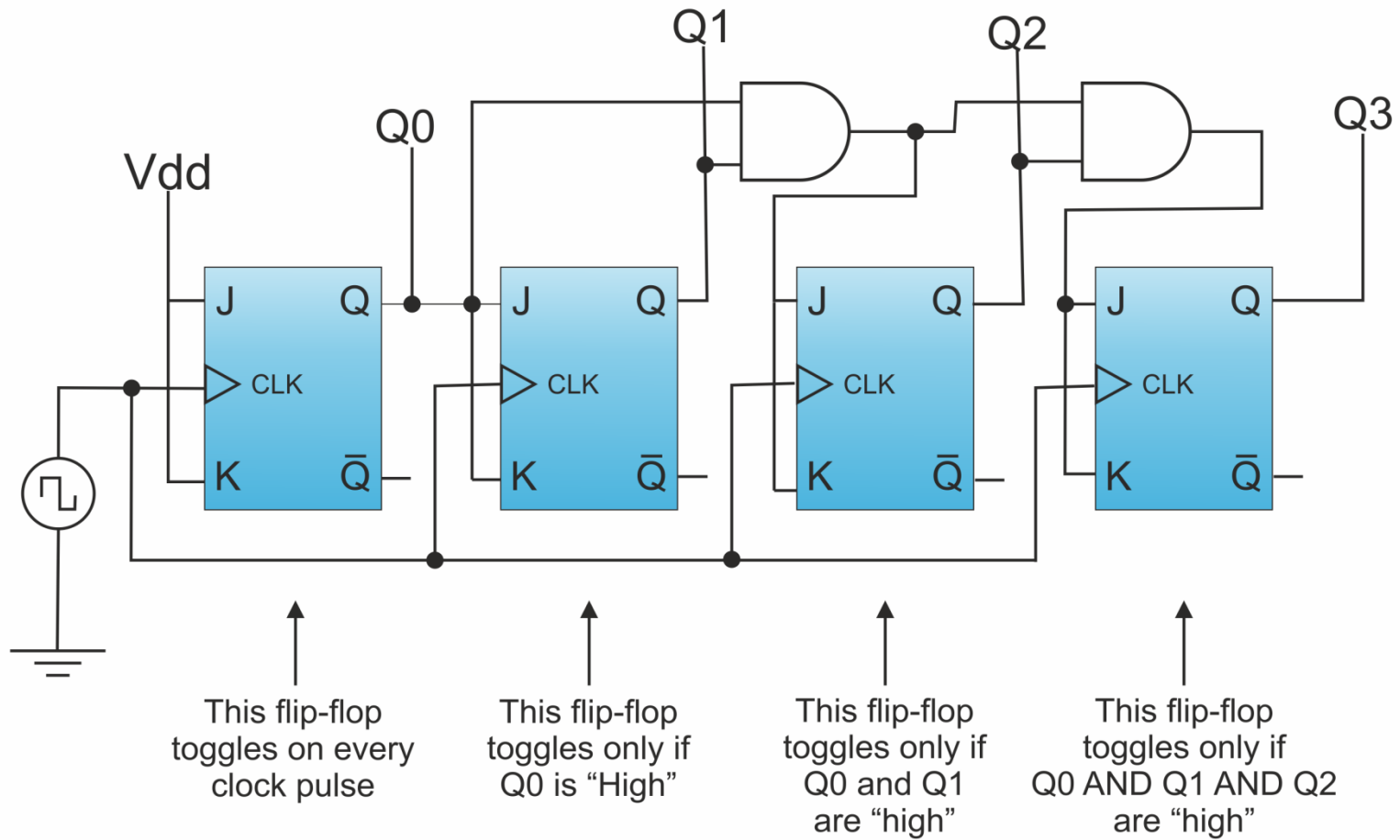


Design of 3 bit Synchronous up counter contd..

- Step 5: Circuit diagram



4-BIT SYNCHRONOUS “UP” COUNTER



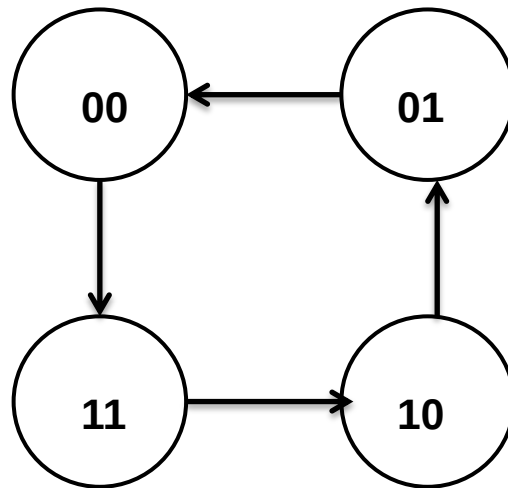
Design of 2 bit Synchronous down counter

- Step 1: No of FFs required=2, Type of FF= T FF
- Step 2: Excitation table of T FF

Present State	Next State	Input
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Design of 2 bit Synchronous down counter cond..

- Step 3:i)State diagram for 2 bit synchronous down counter



Design of 2 bit Synchronous down counter cond..

- Step 3:ii) prepare circuit excitation table

Present State		Next State		FF Inputs	
Q_2	Q_1	Q_{2+1}	Q_{1+1}	T_2	T_1
1	1	1	0	0	1
1	0	0	1	1	1
0	1	0	0	0	1
0	0	1	1	1	1

Design of 2 bit Synchronous down counter cond..

- Step 4: Prepare K map

$Q_2 \backslash Q_1$ K map for T_1

	0	1
0	1	1
1	1	1

$T_1 = 1$

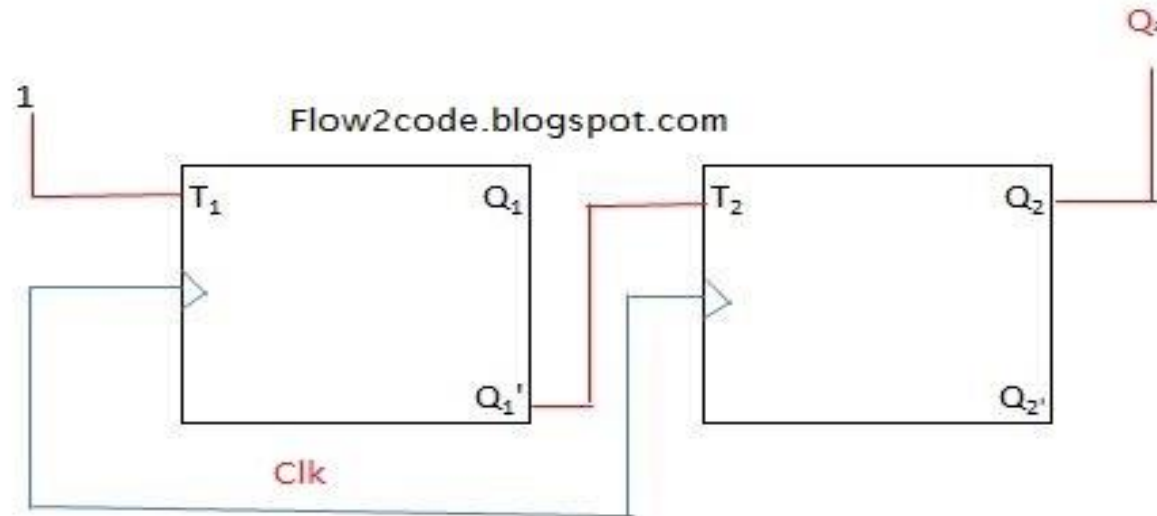
$Q_2 \backslash Q_1$ K map for T_2

	0	1
0	1	0
1	1	0

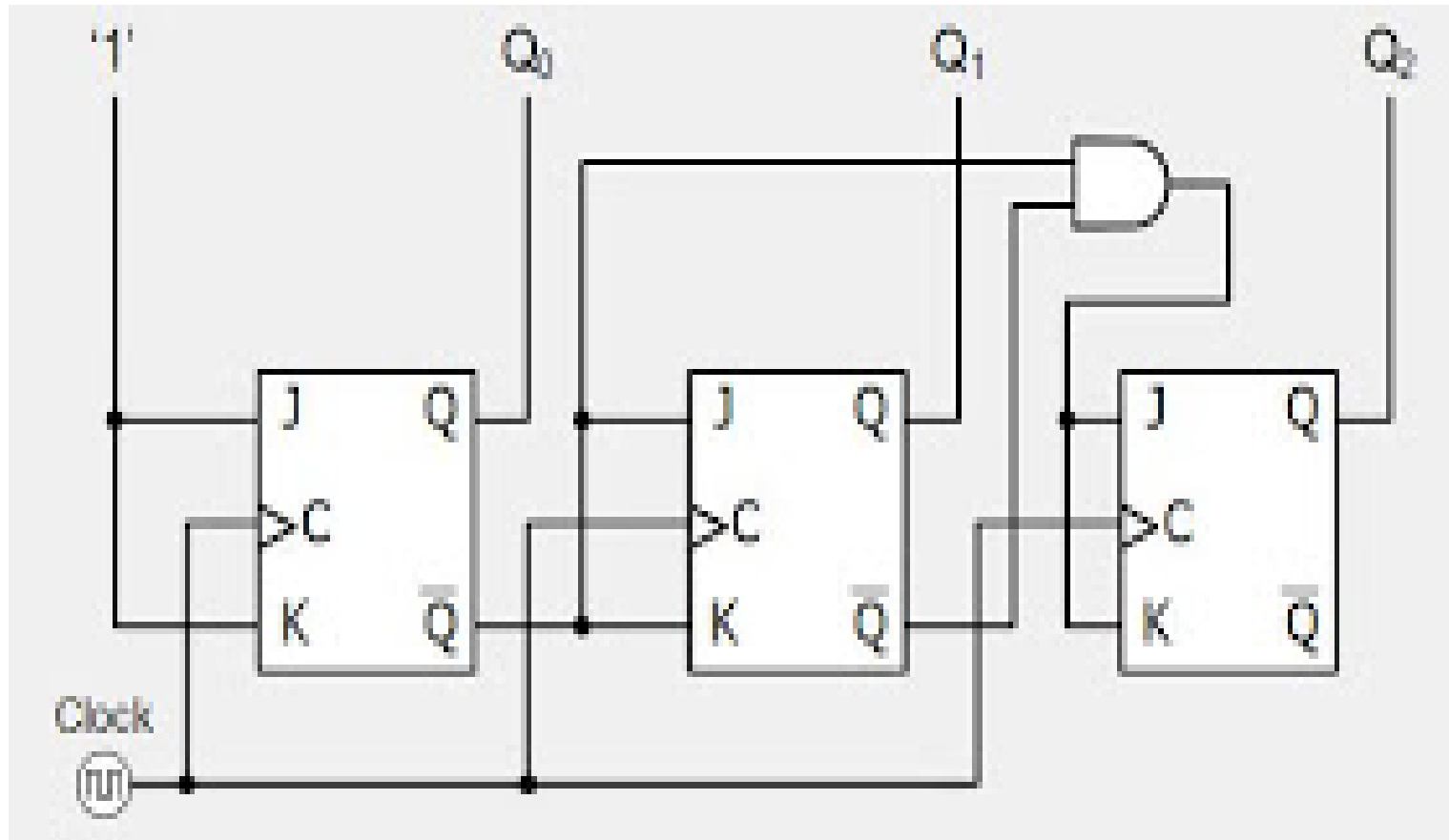
$T_2 = Q_1'$

Design of 2 bit Synchronous down counter cond..

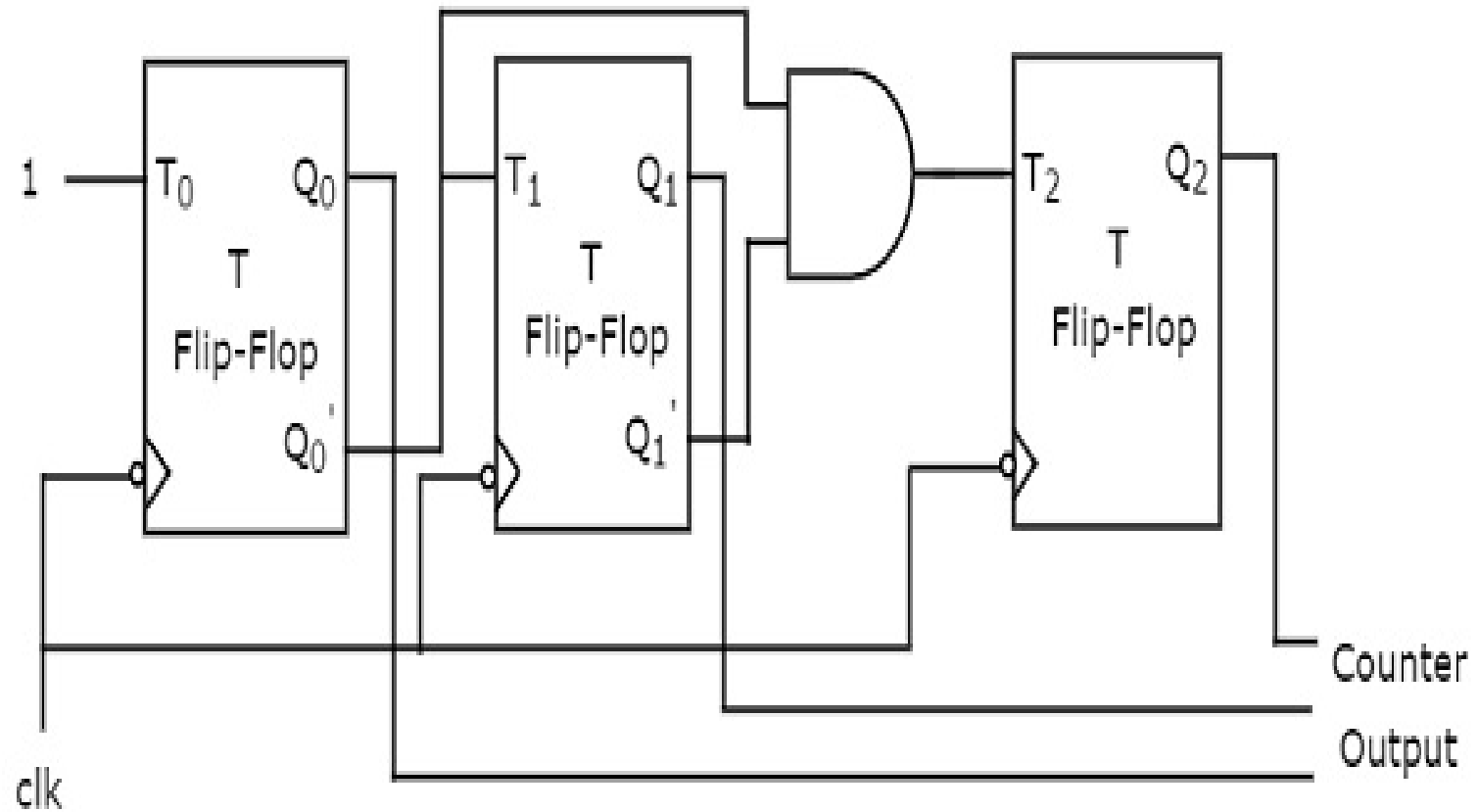
- Step 5: Draw circuit diagram



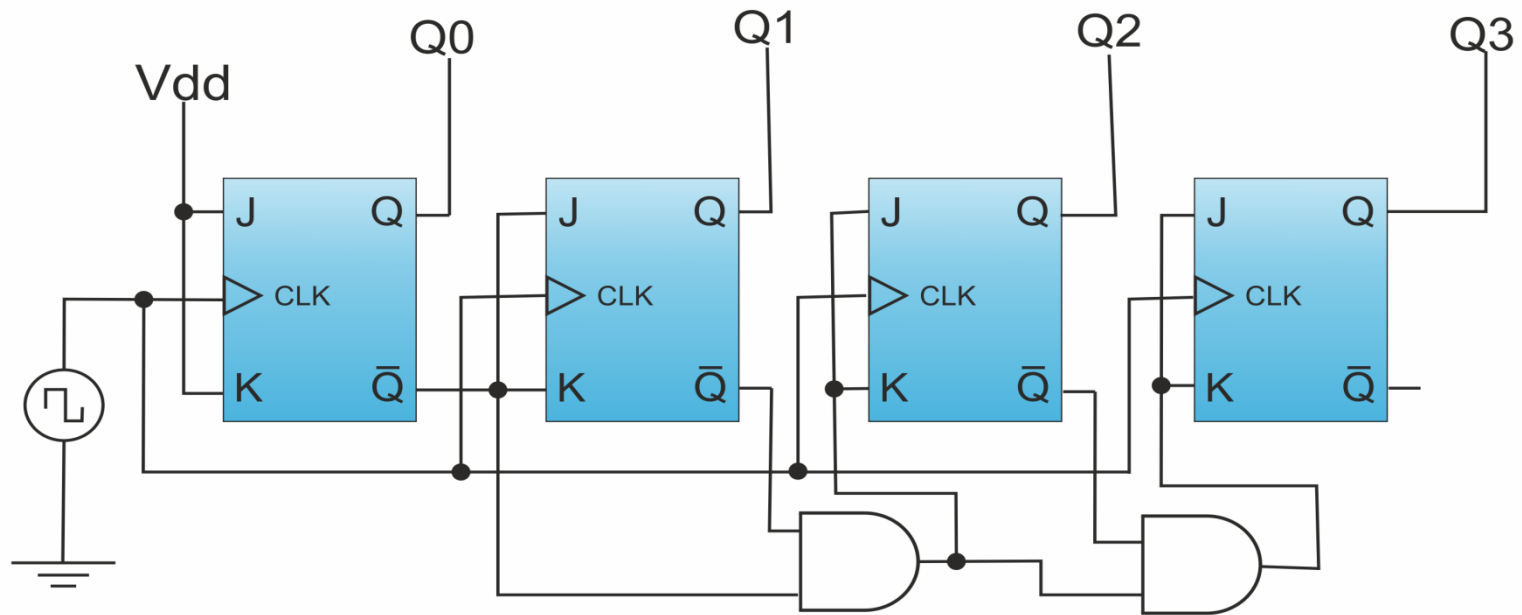
3 bit Synchronous down counter with JK FF



3 bit Synchronous down counter with T FF



4-BIT SYNCHRONOUS “DOWN” COUNTER



↑
This flip-flop
toggles on every
clock pulse

↑
This flip-flop
toggles only if
 $\bar{Q}$ 0 is “High”

↑
This flip-flop
toggles only if
Q0 and $\bar{Q}$ 1
are “high”

↑
This flip-flop
toggles only if
 $\bar{Q}$ 0 AND $\bar{Q}$ 1 AND $\bar{Q}$ 2
are “high”

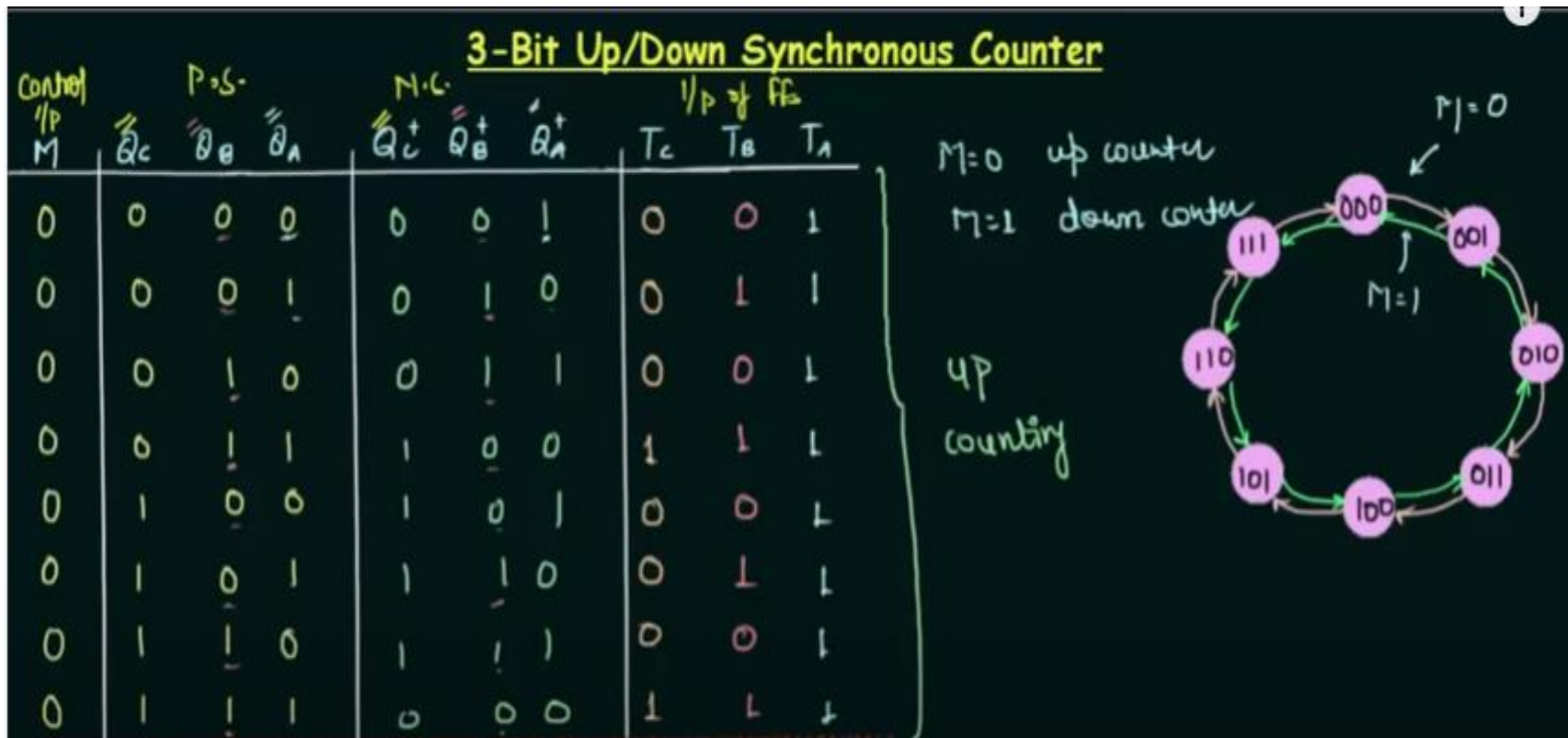
3 bit Synchronous up/down counter

- Step 1 :No of FFs=3,Type of FF- T FF
- Step 2: Prepare excitation table of T FF

Present State	Next State	Input
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

3 bit Synchronous up/down counter cond..

- Step 3i) Prepare state diagram
- Step 3:ii) Prepare circuit excitation table



3 bit Synchronous up/down counter cond..

- Step 3: circuit excitation table for down counting

Control 1/P M	P.S. $\bar{Q}_C$ $\bar{Q}_B$ $\bar{Q}_A$			N.C. $\bar{Q}_C^+$ $\bar{Q}_B^+$ $\bar{Q}_A^+$			1/P of FF T_C T_B T_A		
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

Down counting

3 bit Synchronous up/down counter cond..

- Step 4: K map preparation

for T_C :-

Q_B $M \backslash Q_A$	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	0	0	0
10	1	0	0	0

$T_C = 1 + 1 +$
 $T_C = \overline{M} Q_B Q_A +$
 $M \overline{Q}_B \overline{Q}_A$

for T_B :-

$M \backslash Q_A$	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$T_B = \overline{M} Q_A +$
 $M \overline{Q}_A$
 $= M \oplus Q_A$

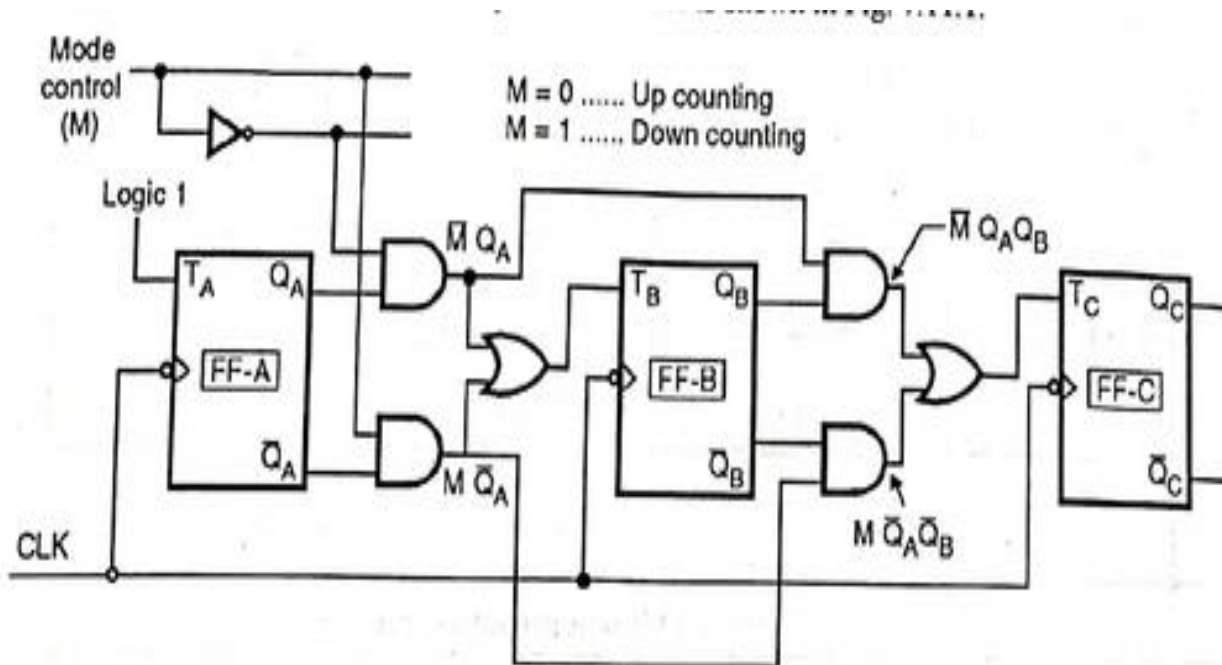
for T_A :-

$Q_B Q_A$ $M \backslash Q_A$	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

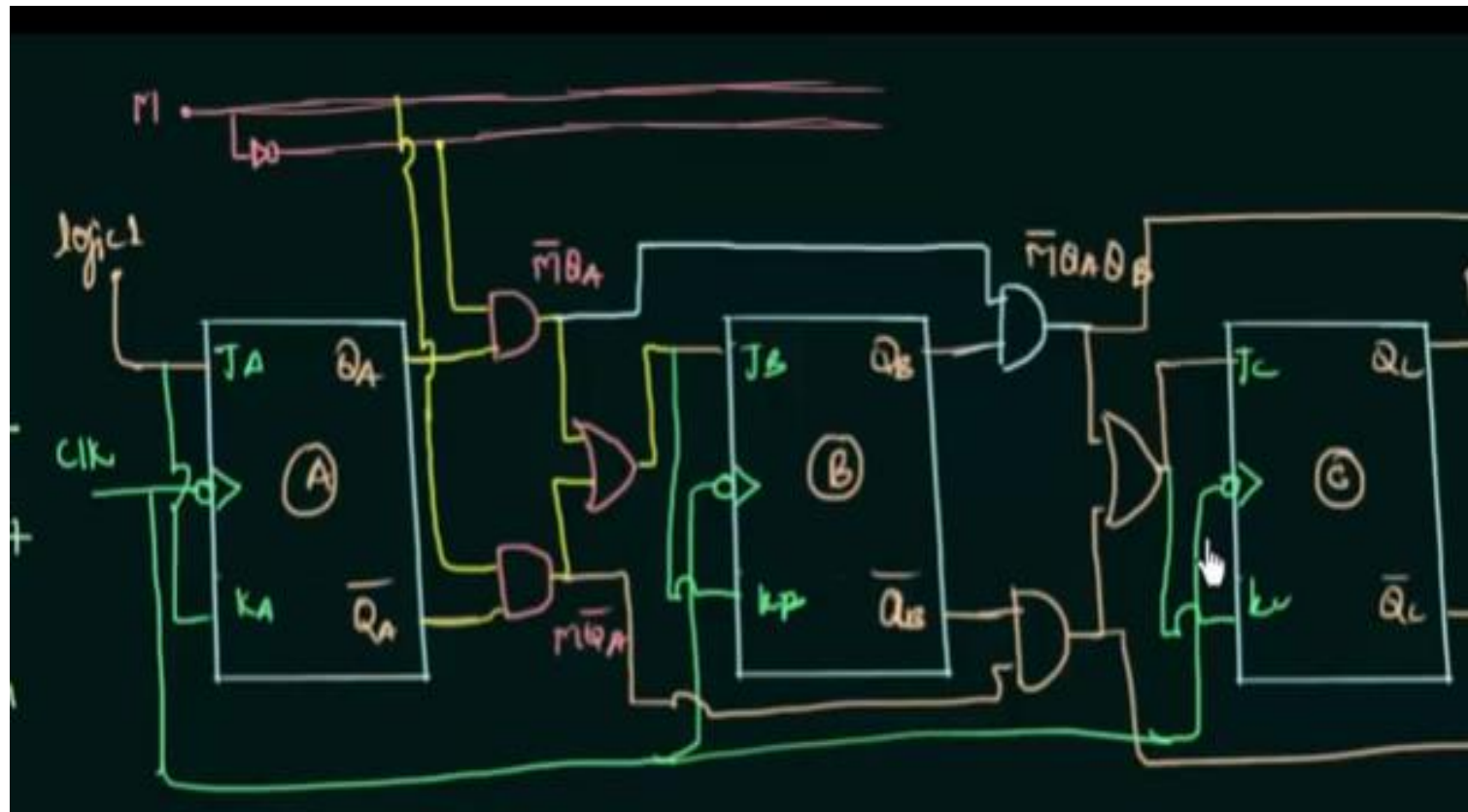
$T_A = 1$

3 bit Synchronous up/down counter cond..

- Step 5: Draw circuit diagram



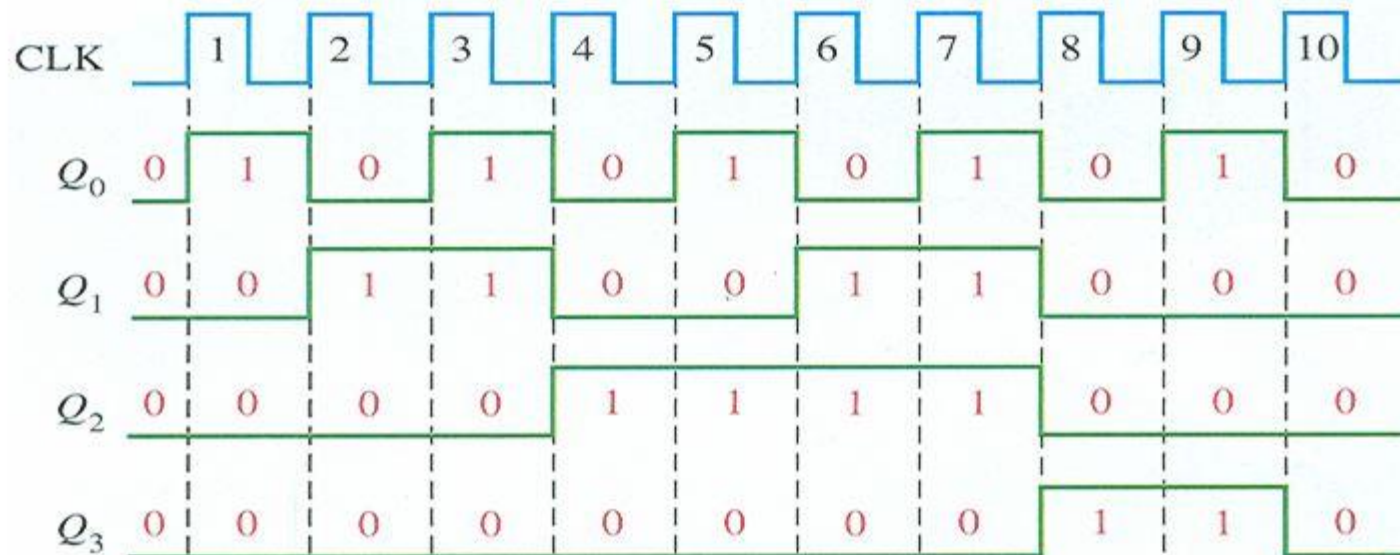
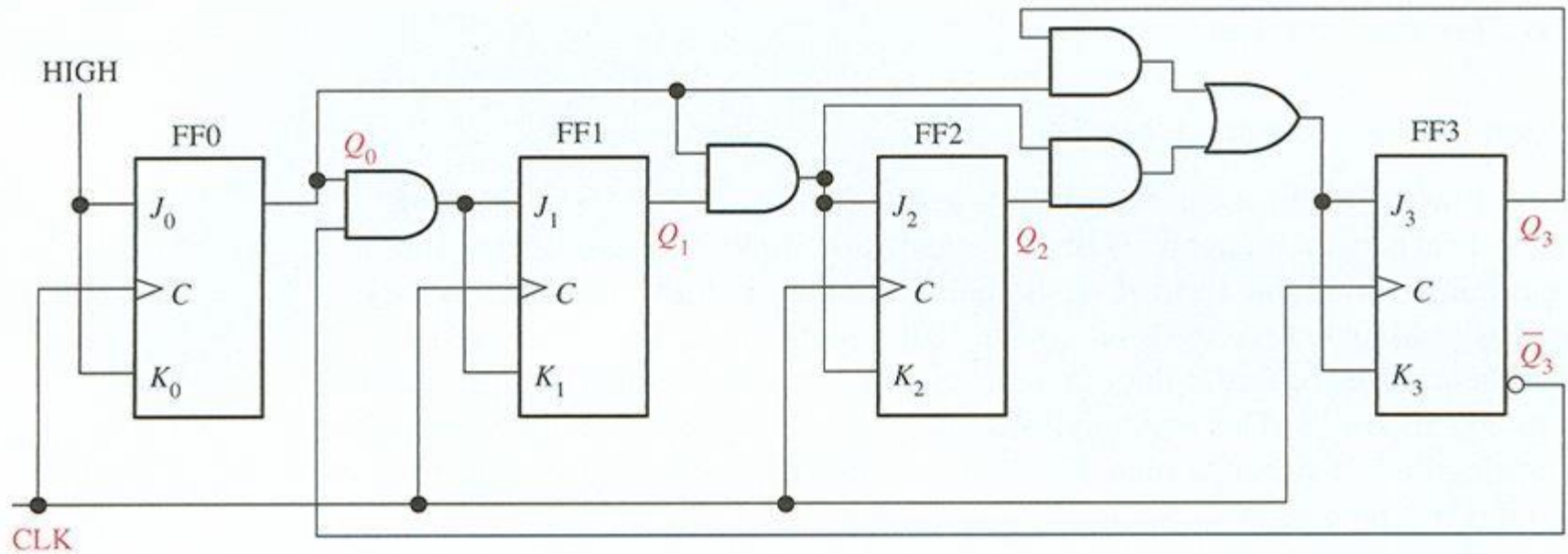
3 bit Synchronous up/down counter using JK FF



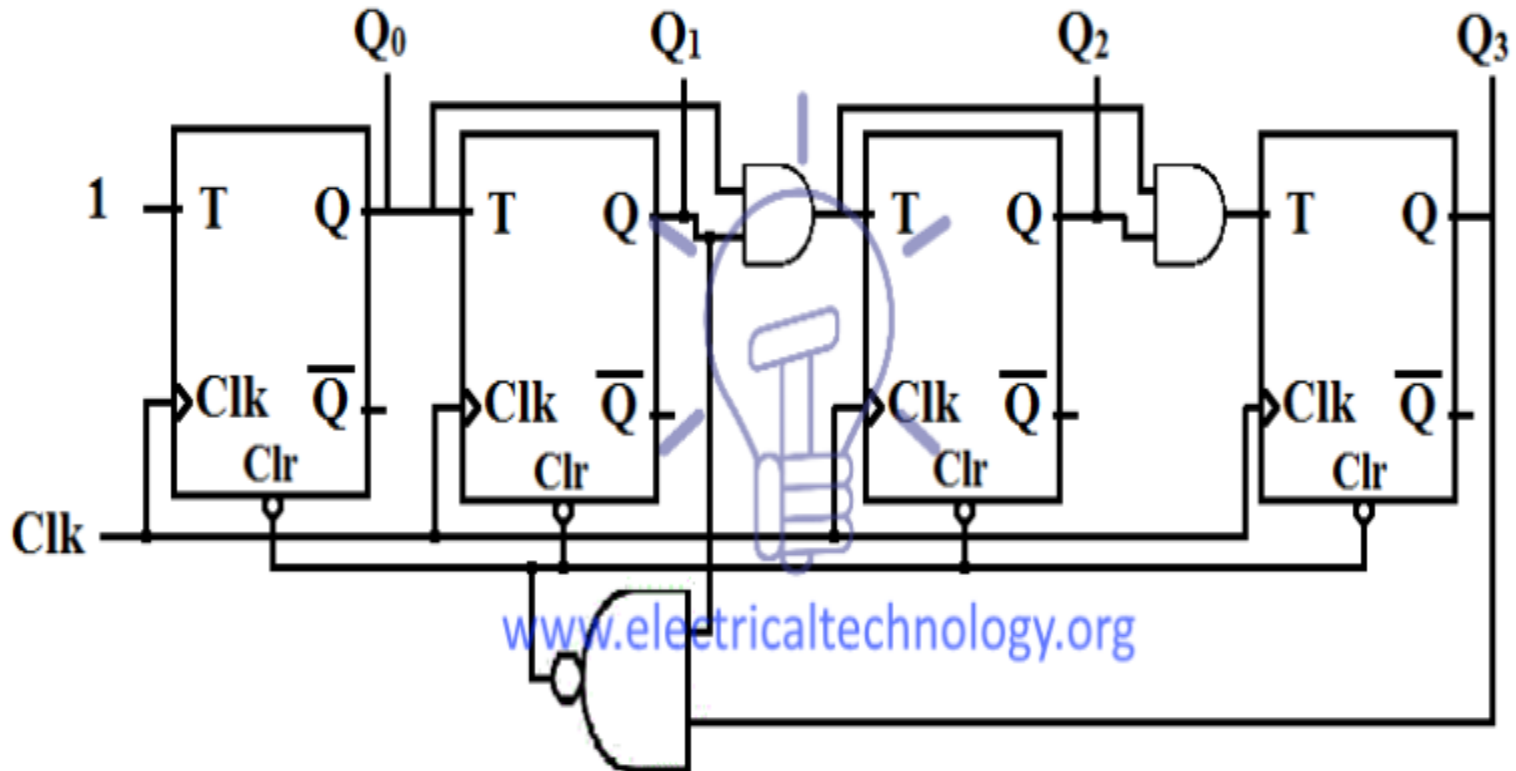
4 bit-Synchronous Decade Counter

- Above circuit is made using Synchronous binary counter, which produces count sequence from 0 to 9. Additional logics are implemented for desired state sequence and to convert this binary counter to decade counter (base 10 numbers, Decimal). When the output reaches count 9 or 1001, the counter will reset to 0000 and again counts up to 1001.
- In the above circuit, AND gates will detect the counting sequence reaches 9 or 1001 and change the state of a third flip-flop from the left, FFC to change its state on the next clock pulse. The counter then resets to 000 and again starts to count until 1001 is reached.

A 4 bit Synchronous Decade Counter



4 bit-Synchronous Decade Counter using T FF



Advantages and Disadvantages of Synchronous Counter

- The **advantages of the Synchronous counter** are as follows-
 - It's easier to design than the Asynchronous counter.
 - It acts simultaneously.
 - No propagation delay associated with it.
 - Count sequence is controlled using logic gates, error chances are lower.
 - Faster operation than the Asynchronous counter.
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- one major **disadvantage** of working with Synchronous counter is that it requires a lot of extra logic to perform.

Applications of Synchronous Counter

- Machine Motion control
- Motor RPM counter
- Rotary Shaft Encoders
- Digital clock or pulse generators.
- Digital Watch and Alarm systems.

Types of Counters | Comparison between Ripple and Synchronous counters



Asynchronous/Ripple Counter

1. Flip flops are connected in such a way that the o/p of first flip flop drives the clock of next flip flop.
2. Flip flops are not clocked simultaneously.
3. Circuit is simple for more number of states.
4. Speed is slow as clock is propagated through number of stages

Synchronous Counter

1. There is no connection between o/p of first flip flop and clock of next flip flop.
2. Flip flops are clocked simultaneously.
3. Circuit becomes complicated as number of states increases.
4. Speed is high as clock is given at a same time.



Thank you !!